

Multiconfiguration Dirac-Hartree-Fock energy levels, oscillator strengths, transition probabilities, hyperfine constants and Landé g -factor of intermediate Rydberg series in neutral argon atom

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Abstract. We computed the energy levels, oscillator strengths f_{ij} , the radiative transition rates A_{ij} , the Landé g -factor, the magnetic dipole moment and the electric quadrupole hyperfine constants of the intermediate Rydberg series $ns [k]_J$ ($4 \leq n \leq 6$), $nd [k]_J$ ($3 \leq n \leq 4$), $np [k]_J$ ($4 \leq n \leq 5$) relative to the ground state $3p^6 \ ^1S_0$ for neutral argon atom spectra. The values are obtained in the framework of the multiconfiguration Dirac-Hartree-Fock (MCDHF) approach. In this approach, Breit interaction, leading quantum electrodynamics (QED) effects and self-energy correction are taken into account. Moreover, these spectroscopic parameters have been calculated for many levels belonging to the configuration $3p^5 4s$, $3p^5 5s$, $3p^5 6s$, $3p^5 3d$, $3p^5 4d$, $3p^5 4p$, $3p^5 5p$ as well as for transitions between levels $3p^5 4s-3p^5 4p$, $3p^5 4p-3p^5 3d$, $3p^5 4p-3p^5 5s$, $3p^5 5s-3p^5 5p$ and $3p^5 5p-3p^5 6s$. The large majority of the lines from the $4p-5s$ and $4p-3d$, $5s-5p$ and $5p-6s$ transition arrays have been observed and the calculations are consistent with the J -file-sum rule. The obtained theoretical values are compared with previous experimental and theoretical data available in the literature. An overall satisfactory agreement is noticed allowing assessing the reliability of our data.

1 Introduction

The argon atomic data such as energy levels, the oscillator strengths f_{ij} , the radiative transition rates A_{ij} , the Landé g -factor, the magnetic dipole moment and the electric quadrupole hyperfine constants are essential in determining the elemental abundance of argon from astronomical observations [1]. They provide benchmarks for theoretical calculations [2], as well as data for applications in plasma physics, the atmospheric sciences and astrophysics [3–6]. Moreover, these data present a crucial step for calculating level populations [7] and spectral lines for non-local thermodynamic equilibrium plasmas [8] and x-ray lasers [9–11].

Built on closed shell $3s^2 3p^6$, the even-parity ground state $3p^6 \ ^1S_0$ yields two odd-parity states $3p^5 \ ^2P_{3/2}$ and $3p^5 \ ^2P_{1/2}$ corresponding to two different values of the total core angular momentum $J_c = \frac{3}{2}$ and $\frac{1}{2}$, respectively. More precisely, two distinctive and adjacent ionization limits corresponding to these two different values of J_c . These limits of ionization are $I_{3/2} = 127\,109.9 \text{ cm}^{-1} \pm 0.1 \text{ cm}^{-1}$ and $I_{1/2} = 128\,541.8 \text{ cm}^{-1} \pm 0.1 \text{ cm}^{-1}$ to which all the corresponding Rydberg series converge [12].

The energy terms of the excited levels belonging to the $3p^5 nl'$ electronic configuration of argon are customarily designated in the jlk -coupling commonly called **Racah coupling** [13]. In this coupling, the quantum number k results from coupling the total angular momentum J_c of the $3p^5$ core with the orbital angular momentum l of the Rydberg electron. The quantum number k is then coupled to the spin, s , of the Rydberg electron to give the total angular momentum J . The fine structure levels belonging to the terms $3p^5 nl'$ are commonly designated by $3p^5 nl [k]_J$.

The energy of the levels belonging to the $3p^5 nl'$ electronic configuration has been measured using various experimental techniques [12, 14–16]. Yoshino [12] used the vacuum spectrograph in the Vacuum-Ultraviolet region in order to access the energy of the odd series $nd [3/2]_1$, $nd [1/2]_1$ and $ns [3/2]_1$ up to values of $n = 58$, 34 and 22, respectively. Salah [14] used laser optogalvanic spectroscopy to measure the energy of the odd-parity $J = 1$ intermediate levels, namely $nd [1/2]_1$ ($3 \leq n \leq 8$), $4d' [3/2]_1$, $6d [3/2]_1$, and $8d [3/2]_1$. For values of principle quantum number $n \leq 18$ for the Rydberg series $nd [3/2]_1$ and $nd [1/2]_1$ and $n = 20$ for $ns [3/2]_1$ these series are perturbed by the levels ns'

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$[1/2]_1$ ($5 \leq n \leq 10$) situated under the first limit of ionization and by the levels nd' $[3/2]_1$ ($3 \leq n \leq 8$) situated over the first limit of ionization where autoionization effects take place. Pellarin and co-workers [15] used a collinear laser spectroscopy technique together with a field ionization detection method to investigate the even-parity spectra of atomic argon. They measured the energy of two hundred and forty one levels belonging to the $3p^5 np$ and $3p^5 nf$ configurations with an absolute accuracy of 0.02 cm^{-1} . Minnhagen [16] successfully measured the energy of the levels issued from the configuration $3p^5 nl$ using grating spectrograph.

The oscillator strength f_{ij} of a transition is defined as the number of classical oscillators per atom for a particular transition. It is also thought of as the fraction of valence electrons likely to participate in the transition of interest. The oscillator strength in neutral argon atom is measured experimentally by different researchers [17–25]. As an example, the oscillator strength for the transition $3p^6 \ ^1S_0$ – $3p^5 4s$ was measured by Copley and Camm [17] and Vallee and co-workers [18] using line broadening, as well as self-absorption by Westerveld and co-workers [19]. Over the years, theoretical studies of the oscillator strength are carried out by researchers [23–32]. Recently, theoretical energy levels and oscillator strengths for neutral argon atom states up to and including $3p^5 \ 3d \ ^1P_1^0$ were computed by Irimia and Fischer [33]. In these calculations, they used Breit-Pauli relativistic corrections and the multiconfiguration Dirac-Hartree-Fock (MCDHF) method.

The transition probability A_{ij} is defined as the probability per unit time that an excited atom in energetic state j makes an allowed (electric dipole) transition to a lower level i . Chornay and co-workers used electron-photon delayed coincidence to measure radiative lifetimes of $3p^5 4s$ excited state [34]. Lately, argon theoretical energies and transition probabilities of $J = 1$ odd and $J = 2$ even states are calculated by Savukov [35]. In his calculation, a combined configuration-interaction and many body perturbation theory were used.

The Landé g -factor is a measure of the magnetic sensitivity of atomic levels. Heretofore, the experimental values for the Landé g -factor of argon neutral atoms are far from being complete. Experimental measurements [14, 36] and theoretical predictions [37] of the Landé g -factor of neutral argon atoms were undertaken by Salah [14], Chenevier and Moskowitz [36] and Aymar and Schweighofer [37] as well as NIST [38]. Salah measured the Landé g -factor for the levels $6d[1/2]_1$, $7d[1/2]_1$, $8d[1/2]_1$ and $9d[1/2]_1$ by using optogalvanique atomic laser spectroscopy. The uncertainty of his results is estimated to be 0.02. On the other side, Chenevier and Moskowitz [37] measured experimentally the Landé g -factor for the levels $3d[1/2]_1$ using optically detected magnetic resonance technique, while the theoretical value of this level, among others, is computed by Aymar and Schweighofer [37]. Both experimental and theoretical values of the Landé g -factor straddle an interesting place in atomic spectroscopy. These values are helpful in a variety of physics applications. They could be used to determine the magnetic dipole A_{MD} and the electric quadrupole B_{EQ} hyperfine constants in cross level experiment [39–41].

The magnetic dipole A_{MD} and the electric quadrupole B_{EQ} hyperfine constants of neutral argon atom are scant.

Nevertheless, these experimental and theoretical studies were restricted to transitions from the ground state to $3p^5 3s$, $3p^5 4p$ and $3p^5 3d$, whereas transitions between excited states and highly excited ones have not been studied by many researchers.

The experimental measurements and theoretical predictions of spectroscopic data of neutral argon atom are intricate by various factors. From an experimental point of view, the electronic transitions from the starting levels to high-lying configuration are difficult to be perceived, because the classical methods of fluorescence detection are inefficient for the highly excited levels. These levels, in general quenched either by collisional ionization, or, spontaneously if they are auto-ionized. Therefore, it is difficult to analyze experimentally their data. From theoretical prediction viewpoint, the calculations are perplexing due to three reasons: the first one is due to the valence orbital in the $3p^5 nl'$ configuration which eliciting strong term dependence, specially, in the $3p^5 nd \ ^1P$ and $3p^5 np \ ^1S$ states. The second one is due to the unfilled $3p$ shell, which inducing spin-orbit coupling of different terms. The last one is due to the strong admixture between the $3p^5(n+1)s$ and $3p^5 nd$ series.

After all, the knowledge of spectroscopic data of high-lying levels of neutral argon atom still suffers from shortcomings. In an effort to fill this lack, also to provide sufficiently accurate new and reliable theoretical spectroscopic data that satisfy their demand in various physical applications, we devote this paper. In this paper, we report and discuss the result of our systematic relativistic calculations of energy levels, oscillator strengths, transition rates, Landé g -factors, the magnetic dipole and the electric quadrupole hyperfine constants of the $ns \ [k]_J$ ($4 \leq n \leq 6$), $nd \ [k]_J$ ($3 \leq n \leq 4$), $np \ [k]_J$ ($4 \leq n \leq 5$) relative to the ground state $3p^6 \ ^1S_0$ for neutral argon atom spectra. These calculations will be carried out by employing a fully relativistic approach based on Dirac equation.

2 Theoretical background and mathematical formulation

2.1 Basic formulae

The energy levels in atomic units of an atom of multi electrons can be calculated from the matrix element of the relativistic Hamiltonian:

$$H = \sum_{i=1}^N H_D(i) + \sum_{i < j}^N \frac{1}{r_{ij}}, \quad (1)$$

where the first term $H_D(i)$, which incorporate the major relativistic effects upon the electrons distribution from the outer shells, represents the Hermitian single-electron Dirac Hamiltonian due to the potential of nuclear charge,

$$H_D = V(r_i) \hat{I} + \left(\hat{\beta} - \hat{I} \right) c^2 + c \hat{\alpha} \cdot \left(\hat{P}_i + \vec{A}(\vec{r}_i) \right), \quad (2)$$

where $c = 137$ in atomic unit, $V(r_i)$ is the electric scalar potential due to the interaction of the electron to the nucleus, $\vec{A}(\vec{r})$ is the magnetic vector potential, $\hat{P} = -i\nabla$ is the electron momentum operator, $\hat{\alpha} = \begin{pmatrix} 0 & \hat{\sigma} \\ \hat{\sigma} & 0 \end{pmatrix}$ is a four-matrix known as Dirac operator and $\hat{\sigma}$ represents the Pauli matrices, and $\hat{\beta} = \begin{pmatrix} \hat{I} & 0 \\ 0 & -\hat{I} \end{pmatrix}$ is a four-matrix known as the Dirac basis where $\hat{I}$ is the unit matrix.

The second term represents the electrostatic interaction of the electrons between each other, and r_{ij} represents the relative position of the electrons.

2.2 Central-field potential and energy levels

In central field, the Dirac Hamiltonian H_D given in eq. (2) becomes

$$H_D(i) = c \hat{\alpha} \cdot \hat{P}_i + \left(\hat{\beta}_i - \hat{I} \right) c^2 - \frac{Z}{r_i} \hat{I}. \quad (3)$$

In this Hamiltonian, the magnetic interaction or retardation effects are excluded but they can, like other radiative effects, be treated as perturbation. However, this Hamiltonian does not commute with the component of the angular-momentum operators $\hat{L}$ and $\hat{L}^2$, whereas it commutes with the total angular momentum $\hat{J}^2$ and $\hat{J}_z$. This imposes a solution to Dirac equation,

$$\left(E + \frac{Z}{r_i} \hat{I} - \left(\hat{\beta}_i - \hat{I} \right) c^2 - c \hat{\alpha} \cdot \hat{P} \right) \psi = 0, \quad (4)$$

of the form ψ_{jm_z} which describes stationary states with a given values of $\hat{J}^2$ and $\hat{J}_z$. Moreover, each state is characterized by its parity. Heretofore, the orbital angular momentum l is not defined; however, the index l will be used to describe the parity of the state. For a given value of j , $l = j \pm 1/2$, one odd and the other even. Taking these conclusions into account, the ansatz stationary eigenfunctions of eq. (10) can be written as a linear combination of the so-called Configuration State Functions (CSF)

$$\Psi(\vec{r}, \sigma) = \sum_{nkm} b_{nkm} \psi_{nkm}(\vec{r}, \sigma), \quad (5)$$

where b_{nkm} are the mixing coefficients that represent $\psi(\vec{r})$ on the orthonormal basis $\{\varphi_{nkm}\}$. These coefficients can be obtained by diagonalizing the $N \times N$ matrix of the total Hamiltonian H . The basis states $\varphi_{nkm}(\vec{r})$; which in relativistic case are spinors of rank 4, are antisymmetric sums of the product of N one-electron Dirac spinors. The one-electron wave function of both the core and the excited one can be written in two-row matrix form as

$$\psi_{nkm}(\theta, \phi, \sigma) = \begin{pmatrix} \varphi_{nkm} \\ \tilde{\varphi}_{nkm} \end{pmatrix} = \frac{1}{r} \begin{pmatrix} i P_{nk}(r) \chi_{km}(\theta, \phi, \sigma) \\ Q_{nk}(r) \chi_{-k,m}(\theta, \phi, \sigma) \end{pmatrix}, \quad (6)$$

where n is the principle quantum number, m is the z -component of the total angular momentum $\vec{J}$, k is the relativistic angular-momentum quantum number commonly called Dirac quantum number. The latter is related to the orbital and total angular momentum l and j , respectively, by

$$k = (l - j)(2j + 1), \quad (7)$$

$i = 1, 2, 3, \dots$, labels the system subshells, $P_{nk}(r)$ and $Q_{nk}(r)$ are the radial functions, and χ_{km} are two component spherical spinors containing the vector coupling of angular momentum and spin. The angular part χ_{km} is a linear combination of spherical harmonics $Y_l^{m-m_s}$ and the spin function S which is a spinor of rank 2:

$$\chi_{km}(\theta, \phi, \sigma) = \sum_{m_s} Y_l^{m-m_s} S^{m_s} \left\langle l \frac{1}{2} m - m_s \mid l \frac{1}{2} j m_j \right\rangle. \quad (8)$$

With this ansatz it is possible to treat all angular- and spin-dependent parts analytically. What remains is the calculation of the two radial functions $P_{nk}(r)$ and $Q_{nk}(r)$. These functions are solutions of the coupled Dirac equation for a local central field $V(r)$. The solution of the Dirac equation will be the aim of the next section.

2.3 The radial functions

The radial functions are the solutions of the coupled Dirac equation for local central field $V(r)$ [42]:

$$\left(\frac{d}{dr} + \frac{\xi}{r}\right) P_{nk}(r) = \alpha \left(\epsilon_{nk} - V_i(r) + \frac{2}{\alpha^2}\right) Q_{nk}(r), \quad (9)$$

$$\left(\frac{d}{dr} - \frac{\xi}{r}\right) Q_{nk}(r) = \alpha (-\epsilon_{nk} - V_i(r)) P_{nk}(r), \quad (10)$$

with

$$\xi = \begin{cases} -\left(j + \frac{1}{2}\right) = -l - 1, & j = l + \frac{1}{2}, \\ \left(j + \frac{1}{2}\right) = l, & j = l - \frac{1}{2}, \end{cases} \quad (11)$$

where α is the fine structure constant, and ϵ_{nk} are the energy levels of an electron in a Coulomb field $V(r)$. These energies are defined by the two quantum numbers n and k . The principal quantum number n is given by

$$n = \acute{n} + |\xi| = \acute{n} + k, \quad |\xi| = k, \quad (12)$$

$$\acute{n} = \begin{cases} 0, 1, 2, 3, \dots, & \xi < 0, \\ 1, 2, 3, \dots, & \xi > 0. \end{cases} \quad (13)$$

On the other hand, the radial functions $P_{nk}(r)$ and $Q_{nk}(r)$ must satisfy the boundary conditions:

$$\left. \begin{matrix} P_{nk}(r) \\ Q_{nk}(r) \end{matrix} \right\} \rightarrow 0, \quad \text{when } \begin{cases} r \rightarrow 0, \\ r \rightarrow \infty. \end{cases} \quad (14)$$

The precedent equation requires an additional normalized condition given by

$$\int_0^\infty (P_{\acute{n}k}(r)P_{nk}(r) + Q_{\acute{n}k}(r)Q_{nk}(r)) dr = \delta_{\acute{n}n}\delta_{kk}, \quad (15)$$

and the local potential $V(r)$ must be of the form

$$V(r) = V^N(r) + V^{e-e}(r), \quad (16)$$

where $V^N(r)$ is the potential due to nuclear charge, given by

$$V^N(r) = \begin{cases} -\frac{Z}{2R_n} \left(3 - \left(\frac{r}{R_n}\right)^2\right), & r \leq R_n, \\ -Z/r, & r > R_n, \end{cases} \quad (17)$$

with Z the atomic number and $R_n = 2.022 \times 10^{-5} A^{1/3}$ denoting an empirically fixed nucleus radius while A is the nucleus mass number; and $V^{e-e}(r)$ is the electron-electron interaction potential. This potential includes the spherically averaged classical potential due to the bound electrons and a local approximation due to the exchange electrostatic interaction. However, $V(r)$ must satisfy the following boundary condition:

$$V(r) \rightarrow \begin{cases} 0, & \text{when } r \rightarrow \infty, \\ -\frac{Z}{r}, & r \rightarrow 0. \end{cases} \quad (18)$$

2.4 The one- and two-electron integral

2.4.1 The one-electron operator

The one-electron operator given by eq. (3) can be written in two-row matrix form as

$$H_D = c \begin{pmatrix} 0 & \hat{\sigma} \\ \hat{\sigma} & 0 \end{pmatrix} \cdot \hat{P} + c^2 \begin{pmatrix} 0 & 0 \\ 0 & -2\hat{I} \end{pmatrix} - \frac{z}{r} \begin{pmatrix} \hat{I} & 0 \\ 0 & \hat{I} \end{pmatrix}. \quad (19)$$

By means of eq. (12), we get

$$\begin{pmatrix} 0 & \hat{\sigma} \\ \hat{\sigma} & \hat{0} \end{pmatrix} \cdot \hat{P} \Psi = \begin{pmatrix} \hat{\sigma} \cdot \hat{P} \tilde{\varphi} \\ \hat{\sigma} \cdot \hat{P} \varphi \end{pmatrix}. \quad (20)$$

By the elementary tensor technique [43], we can show that

$$\hat{\sigma} \cdot \hat{P} \tilde{\varphi}_{nkm} = \left(\frac{1}{r} \frac{\partial}{\partial r} - \frac{k}{r^2} \right) Q_{nk}(r) \chi_{k,m} \quad (21)$$

and

$$\hat{\sigma} \cdot \hat{P} \varphi_{nkm} = -i \left(\frac{1}{r} \frac{\partial}{\partial r} - \frac{k}{r^2} \right) P_{nk}(r) \chi_{-k,m}. \quad (22)$$

By means of eq. (19), we obtain the one-electron operator integral as follows:

$$\begin{aligned} I_a = \langle \Psi | H_D | \Psi \rangle &= \int_0^\infty \left[c P_{nk}(r) \left(\frac{dQ_{nk}(r)}{dr} - \frac{k}{r} Q_{nk}(r) \right) - c Q_{nk}(r) \left(\frac{dP_{nk}(r)}{dr} + \frac{k}{r} P_{nk}(r) \right) \right. \\ &\quad \left. - 2c^2 Q_{nk}^2(r) - \frac{z}{r} \rho_{nk}(r) \right] dr, \end{aligned} \quad (23)$$

where $\rho_{nk}(r)$ is given by

$$\rho_{nk}(r) = P_{nk}^2(r) + Q_{nk}^2(r). \quad (24)$$

2.4.2 The two-electron operator

The two-electrons operator is given by eq. (1) as $\sum_{i<j}^N \frac{1}{r_{ij}}$. In order to calculate the contribution of this operator to the energy, we need first of all to transform this expression in such a way as to separate the radial and angular variables. This separation can be done by expanding $\frac{1}{r_{ij}}$ in a series of Legendre polynomials

$$\frac{1}{r_{ij}} = (r_i^2 + r_j^2 - 2r_i r_j \cos \delta)^{-1/2} = \sum_{k=0}^{\infty} \frac{r_{<}^k}{r_{>}^{k+1}} P_k(\cos \delta). \quad (25)$$

Here the smaller and the greater of the vectors $\vec{r}_i$ and $\vec{r}_j$ are denoted by $r_{<}$ and $r_{>}$, P_k is the Legendre polynomials and δ is the angle between the vectors $\vec{r}_i$ and $\vec{r}_j$, more precisely, between the directions $\theta_1 \varphi_1$ and $\theta_2 \varphi_2$.

By using the theorem of addition of spherical harmonics, eq. (25) can be expressed in terms of tensor product as follows:

$$\frac{1}{r_{ij}} = \sum_{k=0}^{\infty} \frac{r_{<}^k}{r_{>}^{k+1}} C^k(i) \cdot C^k(j) = \sum_q (-)^q C_q^k(i) \cdot C_{-q}^k(j). \quad (26)$$

By means of eq. (26), the two-electron antisymmetrised matrix elements can be written as

$$\left\langle ab \left| \frac{1}{r_{ij}} \right| cd \right\rangle = \sum_{kq} (-)^q R^k(abcd) \langle a | C_q^k | c \rangle \langle b | C_{-q}^k | d \rangle, \quad (27)$$

where a denotes the shell with quantum numbers $n_a k_b$ and the same things for b , c , and d , and the radial part R^k is the relativistic Slater integral given as

$$R^k(abcd) = \iint \frac{r_{<}^k}{r_{>}^{k+1}} [P_a(i)P_c(i) + Q_a(i)Q_c(i)] \times [P_b(j)P_d(j) + Q_b(j)Q_d(j)] dr_i dr_j \quad (28)$$

and the angular part can be expressed in terms of 3j-symbols as follows:

$$\begin{aligned} \langle km | C_q^k | k'm' \rangle &= \langle sljm | C_q^k | sl'j'm' \rangle = \\ &= (-1)^{m-1/2} [(2j+1)(2j'+1)]^{1/2} \begin{pmatrix} j & k & j' \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} j & k & j' \\ -m & q & m' \end{pmatrix} \end{aligned} \quad (29)$$

by assuming that the radial functions are the same within certain shells a , b , c , and d . Here $()$ represents the 3j-symbol.

The total energy with zero-th-order wave functions; due to one- and two-electron operators, can be calculated by means of the formulas driven above. This energy can be written as

$$E = \sum_a N_a I_a + \frac{1}{2} \sum_{a,b,k} N_a [C(abk)F^k(ab) + D(abk)G^k(ab)], \quad (30)$$

where $N_a, N_b \dots$ represent the shells occupation numbers, $F^k(ab) = R^k(abab)$, $G^k(ab) = R^k(abba)$, C and D represent the angular coefficients. These coefficients include summation over the magnetic substates; depend on the state under consideration.

3 The relativistic radiative transition rates A_{fi} and the absorption oscillator strength f_{if}

3.1 The relativistic radiative transition rates A_{fi}

The relativistic radiative transition rates A_{if} per second corresponding to transition from the initial state i to the final state f for an electron outside a closed shell [43, 44] is

$$A_{fi} = 2\alpha\omega_{if} \frac{2j_i + 1}{2L + 1} \left(\begin{matrix} j_f & L & j_i \\ 1/2 & 0 & -1/2 \end{matrix} \right)^2 |\overline{M}_{if}|^2, \quad (31)$$

where the matrix elements $\overline{M}_{if}$ for a relativistic radiative transition of a single electron multipole operator of order L [44]. These matrix elements are either the integral for electron multipole transitions or the integral for the electron magnetic multipole transitions [43].

These radiative transitions are defined as

$$M_{if} = \begin{cases} M_{if}^{(e)} + GM_{if}^{(1)}, & \text{for electron multipole transitions,} \\ M_{if}^{(m)}, & \text{for magnetic multipole transitions,} \end{cases} \quad (32)$$

where

$$M_{if}^{(m)} = (i)^{l+1} \frac{2L+1}{\sqrt{L(L+1)}} (k_f + k_i) I_L^+(\omega_{if}), \quad (33)$$

$$M_{if}^{(e)} = (i)^L \left[\sqrt{\frac{L}{L+1}} [(k_i - k_f) I_{L+1}^+(\omega_{if}) + (L+1) I_{L+1}^-(\omega_{if})] - \sqrt{\frac{L+1}{L}} [(k_i - k_f) I_{L-1}^+(\omega_{if}) - L I_{L-1}^-(\omega_{if})] \right], \quad (34)$$

$$M_{if}^{(1)} = (i)^L [(k_i - k_f) I_{L+1}^+(\omega_{if}) + (L+1) I_{L+1}^-(\omega_{if}) + [(k_i - k_f) I_{L-1}^+(\omega_{if}) L I_{L-1}^-] + (2L+1) J_L] \quad (35)$$

with

$$I_L^\pm = \int_0^\infty j_l \left(\frac{\omega_{if}}{c} r \right) (P_{n_i k_i}(r) Q_{n_f k_f}(r) \pm Q_{n_i k_i}(r) P_{n_f k_f}(r)) dr, \quad (36)$$

$$J_L = \int_0^\infty j_l \left(\frac{\omega_{if}}{c} r \right) (P_{n_i k_i}(r) P_{n_f k_f}(r) + Q_{n_i k_i}(r) Q_{n_f k_f}(r)) dr, \quad (37)$$

where j_l is the spherical Bessel function and G in eq. (32) is the gauge parameter. It takes the value 0 in Coulomb gauge [45, 46] and $\sqrt{\frac{L+1}{L}}$ in Babushkin gauge [47–49]. In the non-relativistic limit, $G = 0$ gives the velocity form of radiation matrix elements, while $G = \sqrt{\frac{L+1}{L}}$ gives the length form.

3.2 The absorption oscillator strength f_{if}

The dimensionless absorption oscillator strength f_{if} for a transition from the initial level i to the final level f are calculated in the single multipole approximation. In this approximation, the interference between different multipoles transitions will not be taken into account, whereas transition rates corresponding to arbitrary multipoles will be taken into account. Under these conditions, the line strength f_{if} for the transition from initial atomic state function Ψ_i to final state function Ψ_f induced by a multipole radiation field operator $\hat{D}_m^{(L)}$ of order L is

$$f_{if} = \frac{\pi c}{(2L+1)\omega_{if}^2} \left| \langle \Psi_i | \hat{D}_m^{(L)} | \Psi_f \rangle \right|^2. \quad (38)$$

The last equation can be written in terms of the reduced matrix elements as

$$f_{if} = \frac{\pi c}{(2L+1)\omega_{if}^2} \sum_{\mu\nu} b_{\mu f} b_{i\nu} \langle n_i k_i || \hat{D}_m^{(L)} || n_f k_f \rangle, \quad (39)$$

where the reduced matrix elements $\langle n_i k_i || \hat{D}_m^{(L)} || n_f k_f \rangle$ are given by

$$\langle n_i k_i | \hat{d}^{(L)} | n_f k_f \rangle = \left(\frac{(2j_i+1)\omega_{if}}{\pi c} \right)^2 (-1)^{j_i-\frac{1}{2}} \begin{pmatrix} j_i & L & j_f \\ 1/2 & 0 & -1/2 \end{pmatrix} |\overline{M}_{if}|. \quad (40)$$

4 Landé g-factor

4.1 Problem set-up

The Hamiltonian $\hat{H}$ describing the interaction of an atom with a uniform magnetic field $\vec{B}$ has the form

$$\hat{H} = \frac{1}{2} \sum_i e \hat{\alpha}_i \cdot (\vec{B} \times \vec{r}_i), \quad (41)$$

where the summation is taken over all atomic i -th electron, e is the electron charge and $\hat{\alpha} = \begin{pmatrix} 0 & \hat{\sigma} \\ \hat{\sigma} & 0 \end{pmatrix}$ is a four-matrix known as Dirac operator and $\hat{\sigma}$ represents the Pauli matrices. For simplicity, we shall drop the index i in the following equations. In terms of the scalar product of magnetic dipole moment spherical tensor of rank 1, $\mu_q^{(1)}$, eq. (41) becomes

$$\hat{H} = \frac{1}{2} \sum \mu_q^{(1)} \cdot \vec{B} \quad (42)$$

with

$$\sum \mu_q^{(1)} = - \sum i e \sqrt{\frac{8\pi}{3}} r \hat{\alpha} \cdot Y_{1q}^{(0)}(\vec{r}), \quad (43)$$

where $Y_{1q}^{(0)}(\vec{r})$ represent the spherical harmonic vector.

The Landé g -factor g_J is defined by the magnetic dipole moment operator $\hat{\mu}$ of an atom in the state $|JM_J\rangle$ as

$$\hat{\mu} = -g_J \mu_B \hat{J}, \quad (44)$$

where μ_B is the Bohr magneton.

Since matrix elements of the Hamiltonian given by eq. (44) can be calculated in the state $|JM_J\rangle$, the matrix elements of the Hamiltonian $\hat{H}$ are

$$\langle \hat{H} \rangle = g_J \mu_B \langle \hat{J} \cdot \vec{B} \rangle, \quad (45)$$

then the Landé g -factor g_J can be written by means of Wigner-Eckart theorem [50] as

$$g_J = \frac{1}{2\mu_B} \frac{\langle J || \sum \mu_q^{(1)} || J \rangle}{\sqrt{J(J+1)}}. \quad (46)$$

On the other hand, the quantum electrodynamics correction imposes a correction on the electron g_s factor [51]. This correction is given by

$$g_s = 2 \left(1 + \frac{\alpha}{2\pi} - 0.3828 \left(\frac{\alpha}{\pi} \right)^2 + \dots \right) \approx 2 \times 1.001160 \quad (47)$$

where $\alpha = \frac{1}{137}$ is the fine-structure constant.

The correction to the electron g_s factor leads to a correction to the relativistic interaction Hamiltonian by an amount

$$\Delta\hat{H} = 0.001160\mu_B\hat{\beta}\hat{\alpha} \cdot \vec{B} \quad (48)$$

and this leads to a correction to Landé g_J factor by an amount

$$\Delta g_J = 0.001160 \frac{\langle J \| \sum \mu_q^{(1)} \| J \rangle}{\sqrt{J(J+1)}}. \quad (49)$$

4.2 Radial matrix elements

The relativistic eigenfunction for a state denoted in Dirac notation by $|\Gamma_J \Pi J M_J\rangle$ can be expanded in JJ -coupled configuration state functions which are eigenfunctions of J^2 , J_z and the parity operator Π in the MCDHF as

$$|\Gamma_J \Pi J M_J\rangle = \sum_r C_r |\gamma_J \Pi J M_J\rangle, \quad (50)$$

where Γ_J and γ_J represent the configuration and any other quantum number required to specify a state, and C_r are the expansion coefficients. On the other hand, the configuration state functions are sums of product of four component spin-orbital function named Dirac wave function. The spin spherical harmonics χ_{km} defined in equation can be written in the LSJ -coupling as

$$\chi_{km} = \sum_q C \left(l \frac{1}{2} j; m - qqm \right) Y_{lm-q}(\vec{r}) \cdot \begin{cases} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, & q = \frac{1}{2}, \\ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, & q = -\frac{1}{2}. \end{cases} \quad (51)$$

Moreover, an angular recoupling computer program [52] is used in order to reduce the matrix elements $\langle J \| \sum \mu_q^{(1)} \| J \rangle$ and $\langle J \| \sum \Delta \mu_q^{(1)} \| J \rangle$ to terms involving single-particle orbitals only

$$\left\langle J \left\| \sum \mu_q^{(1)} \right\| J \right\rangle = \sum_{a,b} D_{ab}^{(1)}(rs) \left\langle n_a k_a \left\| \mu^{(1)} \right\| n_b k_b \right\rangle, \quad (52)$$

$$\left\langle J \left\| \sum \Delta \mu_q^{(1)} \right\| J \right\rangle = \sum_{a,b} D_{ab}^{(1)}(rs) \left\langle n_a k_a \left\| \Delta \mu^{(1)} \right\| n_b k_b \right\rangle, \quad (53)$$

where

$$D_{ab}^{(1)} \equiv - \sum (-1)^{\frac{1}{2}} e \sqrt{\frac{8\pi}{3}} r \hat{\alpha} \cdot Y_{lq}^{(0)}(\vec{r}) \delta_{bb'} \sqrt{2j_a+1} \sqrt{2j_b+1} \cdot (-1)^{a+b+j_b+1} \cdot \left\{ \begin{matrix} b & a & j_a \\ 1 & j_b & a \end{matrix} \right\} \quad (54)$$

and the $\{\}$ denotes the $6j$ -symbol.

However, the single-particle matrix elements can be reduced to angular factors and radial integral:

$$\left\langle n_a k_a \left\| \mu^{(1)} \right\| n_b k_b \right\rangle = -\alpha (k_a + k_b) \left\langle n_a k_a \left\| C^{(1)} \right\| n_b k_b \right\rangle [r]_{n_a k_a n_b k_b}, \quad (55)$$

$$\left\langle n_a k_a \left\| \Delta \mu^{(1)} \right\| n_b k_b \right\rangle = (k_a + k_b - 1) \left\langle n_a k_a \left\| C^{(1)} \right\| n_b k_b \right\rangle [r^{(0)}]_{n_a k_a n_b k_b}, \quad (56)$$

where

$$\left\langle k_a \left\| C^{(1)} \right\| k_b \right\rangle = \begin{cases} (-1)^{j_b-1/2} \sqrt{2j_b+1} \begin{pmatrix} j_a & j_b & 1 \\ 1/2 & -1/2 & 0 \end{pmatrix}, & \text{if } l_a + 1 + l_b \text{ is even,} \\ 0, & \text{if } l_a + 1 + l_b \text{ is odd,} \end{cases} \quad (57)$$

$$[r]_{n_a k_a n_b k_b} = \int_0^\infty r (P_{n_a k_a}(r) Q_{n_b k_b}(r) + Q_{n_a k_a}(r) P_{n_b k_b}(r)) dr, \quad (58)$$

$$[r^{(0)}]_{n_a k_a n_b k_b} = \int_0^\infty r^{(0)} (P_{n_a k_a}(r) P_{n_b k_b}(r) + Q_{n_a k_a}(r) Q_{n_b k_b}(r)) dr. \quad (59)$$

4.3 Hyperfine interaction

In relativistic framework, the hyperfine Hamiltonian is

$$\hat{H}_{hfs} = \sum_i T_n^{(k)}(j) \cdot T_e^{(k)}(i), \quad (60)$$

where $T_n^{(k)}$ and $T_e^{(k)}$ are the spherical tensor operator of rank k , representing the nuclear and electron angular momentum, respectively, j and i designate the various protons and electrons, respectively.

The mean value of $\hat{H}_{hfs}$ in a fine structure state $|JIFM_F\rangle$ is given to first order as

$$\langle \hat{H}_{hfs} \rangle = \sum_i \delta_{M_F M_{\hat{F}}} \delta_{F \hat{F}} (-1)^{F+J+I} \begin{Bmatrix} F & J & I \\ k & I & J \end{Bmatrix} \langle I \| T_n^{(k)} \| I \rangle \cdot \langle J \| T_e^{(k)} \| J \rangle, \quad (61)$$

where I is the nuclear spin and $F = J + I$ is the total angular momentum of the atom.

For the magnetic dipole case, $k = 1$, while the in the electric quadrupole case $k = 2$.

For seek of simplicity, we express the nuclear dipole moment in nuclear magneton μ_n units, the nuclear dipole moment μ_I can be written as

$$\mu_I \mu_n = \langle IM_I = I | T_{n0}^{(k)} | IM_I = I \rangle = \begin{pmatrix} I & k=1 & I \\ -I & 0 & I \end{pmatrix} \langle I \| T_n^{(1)} \| I \rangle. \quad (62)$$

On the other hand, the operator $T_q^{(1)}$ is given by [53, 54]

$$T_q^{(1)} = \sum t_q^{(1)} = - \sum_j ie \sqrt{\frac{8\pi}{3}} \frac{1}{r_j^2} \vec{\alpha}_j \cdot Y_{1q}^{(0)}(\hat{r}_j), \quad (63)$$

where e is the absolute magnitude of the electron charge and j denotes the j -th electron in the atom.

4.3.1 Calculation of the magnetic dipole hyperfine constant A_{MD}

The hyperfine constant A_{MD} is related to the nuclear magnetic moment by

$$A_{MD} = \mu_n \frac{\mu_I}{I} \langle J \| T^{(1)} \| J \rangle (J(J+1)(2J+1))^{-1/2}. \quad (64)$$

By means of eq. (61), the hyperfine energy splitting E_{MD} due to the magnetic dipole is

$$E_{MD} = \frac{A_{MD}}{2} (F(F+1) - I(I+1) - J(J+1)). \quad (65)$$

4.3.2 Calculation of the electric quadrupole constant B_{EQ}

The electric quadrupole constant B_{EQ} is related to the nuclear quadrupole moment Q . The latter is given by

$$eQ = 2 \langle I, M_I = I | Q_0^{(2)} | I, M_I = I \rangle = \begin{pmatrix} I & k=2 & I \\ -I & 0 & I \end{pmatrix} 2 \langle I \| Q_0^{(2)} \| I \rangle. \quad (66)$$

Since e is the charge and Q is the operator for the proton, then eQ represent the charge repartition in the nucleus.

On the other hand, the operator $T_q^{(2)}$ is given by [55] as

$$T_q^{(2)} = \sum t_q^{(2)} = \sum_j \frac{-e}{r_j^3} \sqrt{\frac{\pi}{(2k+1)}} Y_{qk}. \quad (67)$$

Here, the summation is carried out over all electrons of the atom.

The electric quadrupole hyperfine constant becomes

$$B_{EQ} = 2eQ \left[\frac{2J(2J-1)}{(2J+1)(2J+2)(2J+3)} \right]^{1/2} \langle J \| T^{(2)} \| J \rangle. \quad (68)$$

By means of eq. (68), the hyperfine energy splitting E_{EQ} due to the electric quadrupole is

$$E_{EQ} = \frac{B_{EQ}}{8} \frac{3C(C+1) - 4J(J+1)I(I+1)}{J(2J-1)I(2I-1)}, \quad (69)$$

$$C = F(F+1) - I(I+1) - J(J+1). \quad (70)$$

5 Evaluation of uncertainty in theoretical results

The evaluation of uncertainties associated with the theoretical results is mandatory for comparison with the experimental results. Uncertainty is shown to act as a judge of the acceptability of the computability of theoretical predictions with experimental ones.

The uncertainty of transition energy in spin-orbit coupling (LS -coupling) is given by [56] as

$$\delta E = \frac{|E_{\text{calculated}} - E_{\text{experimental}}|}{E_{\text{experimental}}}. \quad (71)$$

Consequently, the uncertainty in the theoretical values of the Landé g -factor g can be given by

$$\delta g = \frac{|g_{\text{calculated}} - g_{\text{experimental}}|}{g_{\text{experimental}}} \quad (72)$$

However, the uncertainty of the computed transition rates A_{fi} is given by [57] as

$$\delta \tilde{A}_{fi} = \frac{|\dot{A}_{fi}(l) - \dot{A}_{fi}(v)|}{\max(\dot{A}_{fi}(l), \dot{A}_{fi}(v))}, \quad (73)$$

where $\dot{A}_{fi}(l)$ and $\dot{A}_{fi}(v)$ are the energy scaled-length and velocity, respectively. These scaled are estimated by [57] as

$$\dot{A}_{fi}(l) = A_{fi}(l) \left(\frac{E_{\text{experimental}}}{E_{\text{calculated}}} \right)^3, \quad (74)$$

$$\dot{A}_{fi}(v) = A_{fi}(v) \left(\frac{E_{\text{experimental}}}{E_{\text{calculated}}} \right). \quad (75)$$

In the relativistic framework, the validity of eqs. (74) and (75) is implausible. Therefore the accuracy in the computed values of the oscillator strengths will be judged by comparing them with the experimental and theoretical data available in the literature.

6 Numerical calculations and discussions

6.1 Computational procedure

The above equations are plugged in a power full **General Relativistic Structure Package** code called **GRASP2K** [58]. It is an integrated software package to predict various atomic spectroscopy parameters, such as atomic energy levels, oscillator strengths, radiative decay rates, hyperfine structure constants, Landé g -factors, and specific mass shift parameters. These parameters are predicted by using a MCDHF approach. As an approximation to improve the numerical results, the core $1s^2 2s^2 2p^6$ was assumed to be inactive. Under this assumption, an expansion of configuration state functions (CSFs) derived from an orbital set was used. Moreover, the restrictive active space method was used to generate CSFs. These CSFs are linear combinations of Slater determinants of atomic orbitals, and are simultaneous eigenfunctions of the atomic electronic total angular-momentum operator, J , and the atomic parity operator. A linear combination of these CSFs is employed in order to generate an approximate atomic state functions (ASFs).

The CSFs were generated by excitation of orbitals occurring in the reference set of configurations to a set of orbitals. In this excitation, the set of orbitals $\{1s, 2s, 2p, 3s, 3p, 3d\}$ and $\{1s, 2s, 2p, \dots, 4s, 4p, 4d, 4f\}$ are denoted as the $n = 3$ and 4 orbital set, respectively. The orbitals are determined by optimization of an energy expression for an LS term.

A step wise increasing of the orbital set was helpful to control the convergence of our numerical results. In the MCDHF approach, orbitals are determined by optimization of an energy expression for an LS term. As soon as the radial functions are determined, relativistic configuration interaction (RCI) calculations were carried out to determine CSF expansion coefficients. These coefficients were determined by diagonalizing the Hamiltonian matrix that included the frequency-dependent Breit interaction, vacuum polarization and self-energy correction.

The radial functions and expansion coefficients were enhanced to self-consistency using the multiconfiguration self-consistent field (MCSCF) procedure [59]. The corresponding eigenvalues and eigenvectors are determined by means of the iterative Davidson method [60]. After that, we used these eigenvectors in order to calculate the required allowed transition data between levels of odd and even parities. Selection rules for electric-dipole transitions in the jk coupling scheme are: $\Delta J = 0, \pm 1$, $\Delta j = 0$, $\Delta k = 0, \pm 1$ and $\Delta l = \pm 1$. However, these rules are not entirely respected, since transitions with change of the ionic core as well as transitions with $\Delta l = 3$ are detected [15]. These transitions are allowed because of the small mixing of the initial $4s [3/2]_2$ level with some levels of the $3p^5 3d$ configuration. A multiconfigurational computation fulfilled in the frame of central-field approximation [61] has shown that the $4s [3/2]_2$ is mixed with $3d [3/2]_2$ and $3d' [3/2]_2$ levels but is not coupled to the $3d [5/2]_2$ and $3d' [5/2]_2$ levels. Under these state of affairs, the allowed transitions are restricted to the selection rules $\Delta J = \pm 1$, $\Delta k = 0, \pm 1$. Moreover, the electric dipole transition rule requires that the parity of initial and final state must change for an allowed transition.

For the computation of the energy levels of neutral argon atom, a list of CSFs was obtained based on the notion of excitations from subshells to an active set of orbitals, *i.e.* the active set method. More precisely, for each specified parity and J symmetry, the list of CSFs is generated, by taking into account the excitation from a given single reference configuration to a set of definite relativistic orbitals. By keeping all orbital subshells active, a permitted single excitation from $1s^2 2s^2 3s^2 3p^5 ({}^2P^0)$ reference configuration, to each one of nl relativistic orbitals, is considered. In our study, the principle quantum number, n goes from $3 \leq n \leq 7$ and l is corresponding to all s, p, d and f orbitals under consideration. By taking the stability of the code into account, the calculations were done by separating each optimization stage corresponding to a given n and l for the outer shell orbital from another different optimization stage corresponding to n' and l' for the outer shell orbital. After the stage of relativistic state configuration function RSCF calculations, the RCI calculations stage is carried out. In the latter stage, both Breit interaction, quantum electrodynamics (QED) effects and self-energy correction were taken into account. Moreover, the expansions for the RCI calculations were obtained by allowing all single excitations to the $3 \leq n \leq 7$ basis orbitals from the reference configurations, with keeping all subshells orbitals active.

The hyperfine structure constants and Landé g -factors are calculated to first-order hyperfine interaction in the frame of MCDHF method. Hence, the distortions of electron shell by nuclear moments are not taken into account in the calculations [62, 63]. The selected wave functions $|J, M_J\rangle$ used in MCDHF consist of all possible jj -coupled states that come up from the $3p^5 3d$ electronic configurations. By means of these wave functions, we built the eigenstates $|I, J, F, M_F\rangle$ of the hyperfine Hamiltonian. Here I and J being the total angular momentum of the nucleus and the electron state, respectively, and $\vec{F} = \vec{I} + \vec{J}$.

More details of computational approach can be found in [64–66].

6.2 Computation of energy levels and oscillator strengths in neutral argon atom

The energy levels of the intermediate Rydberg series $ns [k]_J$ ($4 \leq n \leq 6$), $nd [k]_J$ ($3 \leq n \leq 4$), $np [k]_J$ ($4 \leq n \leq 5$) relative to the ground state $3p^6 {}^1S_0$ of neutral argon atom are numerically calculated and illustrated in table 1. In order to assess the reliability of our data, we compare them with the experimental values measured precisely with uncertainty of 0.05 cm^{-1} by Minnhagen [16]. Within the uncertainty in our theoretical results, our results are in satisfactory agreement with Minnhagen results. However, the differences in the experimental and our computed energies are due to two factors: the first factor is due to the calculations of the central potential for radial orbitals and recoupling schemes of angular parts, the second factor is due to the degeneracy of the 1D and 3D terms of $p^5 nd$ configuration, the relativistic corrections induced strong admixture of terms in certain levels. In order to guarantee the reliability of our dissection, the results admixture terms were reproduced from the MCDHF wave functions. These wave functions were formed by a transformation from the jj to LSJ coupling. Since the LSJ coupling admixtures terms of the same angular momentum J , the $3p^5 nd {}^1P_1$ and 3P_1 levels exhibit a very feeble admixture of terms, whereas the 1F_3 , 3F_3 and the 3D_3 levels exhibit a strong admixture of the corresponding terms.

Recently, Charlotte Froese Fischer and co-workers [67] computed the energy of the levels belonging to the series $ns [k]_J$ ($4 \leq n \leq 5$), $3d [k]_J$ and $4p [k]_J$ of neutral argon atom. Their results are tabulated in table 1. Their data are computed in the frame of multiconfiguration Hartree-Fock (MCHF), and also multiconfiguration Dirac-Hartree-Fock (MCDHF). In MCHF methods, the relativistic effects through the Breit-Pauli Hamiltonian are taken into account.

Table 1. Energy data (cm^{-1}) with their uncertainties for Ar I relative to the ground state $3p^6\ ^1S_0$.

Configuration	Designation	LS-Term	Experimental energy	Computed energy	
			$E_{\text{exp. [16]}}$	This work	$E_{\text{Comp. [67]}}$
4s	4s[3/2] ₂	3P_2	93143.76	$93162.20 \pm 1.98e^{-4}$	92758.95
	4s[3/2] ₁	3P_1	93750.5978	$93768.17 \pm 1.87e^{-4}$	93346.06
	4s[1/2] ₀	3P_0	94553.6652	$94602.75 \pm 5.19e^{-4}$	94064.43
	4s'[1/2] ₁	1P_1	95399.8276	$95409.35 \pm 9.98e^{-5}$	94963.42
3d	3d[1/2] ₀	3P_0	111667.766	$112440.54 \pm 6.92e^{-3}$	111601.68
	3d[1/2] ₁	3P_1	111818.028	$112569.47 \pm 6.98e^{-3}$	111733.94
	3d[7/2] ₄	3F_4	112750.153	$112880.93 \pm 1.16e^{-3}$	112017.90
	3d[7/2] ₃	3F_3	113020.355	$113053.79 \pm 2.95e^{-4}$	112586.49
	3d[3/2] ₂	3P_2	112138.924	$112829.07 \pm 6.15e^{-3}$	112017.90
	3d[3/2] ₁	3D_1	114147.732	$114160.68 \pm 1.13e^{-4}$	113755.20
	3d[5/2] ₂	$^3F_2^\dagger$	113425.964	$113407.88 \pm 1.59e^{-4}$	113169.05
	3d[5/2] ₃	$^1F_3^\dagger$	113716.555	$113515.12 \pm 1.77e^{-3}$	113388.64
	3d[5/2] ₂	$^1D_2^\dagger$	114640.992	$114608.42 \pm 2.84e^{-4}$	114299.65
	3d'[5/2] ₃	$^3D_3^\dagger$	114821.939	$114703.36 \pm 1.03e^{-3}$	114441.85
	3d'[3/2] ₂	3D_2	114805.135	$114735.77 \pm 6.04e^{-4}$	114435.11
	3d'[3/2] ₁	1P_1	115366.866	$115718.48 \pm 3.04e^{-3}$	114908.04
	$^\dagger\ ^3F_2$ couples with 1D_2 , 3D_3 couples with 1F_3				
4p	4p[1/2] ₁	3S_1	104102.099	$104396.67 \pm 2.84e^{-3}$	103949.93
	4p[5/2] ₃	3D_3	105462.759	$105485.41 \pm 2.14e^{-4}$	105061.78
	4p[5/2] ₂	$^3D_2^\dagger$	105617.27	$105633.95 \pm 1.58e^{-4}$	105205.50
	4p[3/2] ₁	$^3D_1^\dagger$	106087.2598	$106112.57 \pm 2.38e^{-4}$	105640.46
	4p[3/2] ₂	$^3P_2^\dagger$	106237.5518	$106226.97 \pm 9.96e^{-4}$	105774.35
	4p[1/2] ₀	3P_0	107054.272	$107070.34 \pm 1.50e^{-4}$	106631.79
	4p'[3/2] ₁	$^1P_1^\dagger$	107131.708	$107176.56 \pm 4.18e^{-4}$	106618.21
	4p'[3/2] ₂	1D_2	107289.7001	$107302.56 \pm 1.12e^{-4}$	106754.22
	4p'[1/2] ₁	3P_1	107496.4166	$107527.15 \pm 1.86e^{-4}$	106961.42
	4p'[1/2] ₀	1S_0	108722.6194	$108901.09 \pm 1.64e^{-3}$	108753.29
	$^\dagger\ ^3D_1$ couples with 1P_1 .				
5s	5s[3/2] ₂	3P_2	113468.473	$112977.04 \pm 4.33e^{-3}$	112916.48
	5s[3/2] ₁	3P_1	113643.260	$114554.20 \pm 8.01e^{-3}$	114328.61
	5s[1/2] ₀	3P_0	114861.635	$114421.87 \pm 3.82e^{-3}$	114211.26
	5s'[1/2] ₁	1P_1	114975.019	$113186.32 \pm 1.55e^{-2}$	113082.60
4d	4d[1/2] ₀	3P_0	118512.197	$118381.45 \pm 1.10e^{-3}$	
	4d[1/2] ₁	3P_1	118651.395	$118472.16 \pm 1.51e^{-3}$	
	4d[7/2] ₄	3F_4	119023.648	$118607.51 \pm 3.49e^{-3}$	
	4d[7/2] ₃	3F_3	119212.870	$118722.39 \pm 4.11e^{-3}$	
	4d[3/2] ₂	3P_2	118906.611	$118632.00 \pm 2.31e^{-3}$	
	4d[3/2] ₁	3D_1	119847.72	$119444.00 \pm 3.37e^{-3}$	
	4d[5/2] ₂	3F_2	119444.83	$120170.29 \pm 6.07e^{-3}$	
	4d[5/2] ₃	1F_3	119566.03	$120245.28 \pm 5.68e^{-3}$	
	4d'[5/2] ₂	1D_2	120619.016	$118906.42 \pm 1.42e^{-2}$	
	4d'[5/2] ₃	3D_3	120753.471	$118959.83 \pm 1.48e^{-2}$	
	4d'[3/2] ₂	3D_2	120600.895	$120212.44 \pm 3.22e^{-3}$	
	4d'[3/2] ₁	1P_1	121011.919	$120884.19 \pm 1.05e^{-3}$	

Table 1. Continued.

Configuration	Designation	LS-Term	Experimental energy	Computed energy	
			$E_{\text{exp.}}$ [16]	This work	$E_{\text{Comp.}}$ [67]
5p	5p[1/2] ₁	$^3\text{S}_1$	116659.9934	$115959.95 \pm 6.00\text{e}^{-3}$	
	5p[5/2] ₃	$^3\text{D}_3$	116942.7542	$116209.39 \pm 6.27\text{e}^{-3}$	
	5p[5/2] ₂	$^3\text{D}_2^\dagger$	116990.3259	$116265.40 \pm 6.20\text{e}^{-3}$	
	5p[3/2] ₁	$^3\text{D}_1$	117151.3264	$117696.58 \pm 4.65\text{e}^{-3}$	
	5p[3/2] ₂	$^3\text{P}_2$	117183.5901	$116447.51 \pm 6.28\text{e}^{-3}$	
	5p[1/2] ₀	$^3\text{P}_0$	117562.9533	$116851.36 \pm 6.05\text{e}^{-3}$	
	5p'[3/2] ₁	$^1\text{P}_1$	118407.4303	$116424.78 \pm 1.67\text{e}^{-2}$	
	5p'[3/2] ₂	$^1\text{D}_2^\dagger$	118469.0508	$117750.48 \pm 6.06\text{e}^{-3}$	
	5p'[1/2] ₁	$^3\text{P}_1$	118459.5975	$117760.57 \pm 5.90\text{e}^{-3}$	
	5p'[1/2] ₀	$^1\text{S}_0$	118870.9170	$118212.63 \pm 5.54\text{e}^{-3}$	
$^\dagger \ ^3\text{D}_2$ couples with $^1\text{D}_2$					
6s	6s[3/2] ₂	$^3\text{P}_2$	119683.0821	$118859.94 \pm 6.87\text{e}^{-3}$	
	6s[3/2] ₁	$^3\text{P}_1$	119760.1725	$120353.86 \pm 4.96\text{e}^{-3}$	
	6s[1/2] ₀	$^3\text{P}_0$	121096.605	$120305.76 \pm 6.53\text{e}^{-3}$	
	6s'[1/2] ₁	$^1\text{P}_1$	121161.3135	$118948.13 \pm 1.82\text{e}^{-2}$	

However, the orbit-orbit interaction is ignored by [67]. The slight differences between our results and those reported by Charlotte Froese Fischer might due to the approximations that were carried out in her calculations. These approximations, among other, are in connection with the determination of the radial wave functions in jj -coupling and the core correlation model.

The comparison of our results with those observed experimentally [16] or computed theoretically [67] ensure in some sense, the consistency of our results.

In the interpretation of our results concerning the computed energy, we observed that the following levels are strongly coupled. The levels 3F_2 and 3D_3 belonging to the 3d $[k]_J$ series is strongly coupled with 1D_2 and 3D_3 belonging to the same series, respectively. Similar couplings are observed in the 4p $[k]_J$ and 5p $[k]_J$ series. In the former, the level 3D_1 and 3P_2 coupled with the levels 1P_1 and 3D_2 . In the latter series, the level 3D_2 couple with 1D_2 .

Absolutely, the coupling between these levels is due to the spin-orbit coupling which admixture sates of the same angular momentum J , which issue from the same configuration.

Indeed, interaction between two series which possess the same total angular momentum J and parity will arise. Therefore, more attention should be paid in order to monitor the reliability of the results concerning the prediction of the energy levels belonging to such series.

Based on the accurate position of our levels, the corresponding oscillator strengths f_{ij} were calculated in the length ($f_{ij}(l)$) and velocity ($f_{ij}(v)$) forms of the electric dipole operator forms in Babushkin and Coulomb gauge, respectively. The obtained results are illustrated in table 2. In this table g_i and g_k are the statistical weight of the levels i and k , respectively. The disparity between length and velocity values of the oscillator strengths is expected in relativistic calculation whereas the agreement is better in the non-relativistic calculations. In the relativistic calculation, the optical oscillator strength should be the mean value of the oscillator strength in length and velocity form. Therefore, the comparison of these forms in relativistic case cannot be used to provide an idea of the accuracy of the wave functions.

Since the transition from the ground state to the levels 4p is not allowed by parity selection rule, we calculated all the transitions $3p^5(^2P)4s-3p^5(^2P)4p$, $3p^5(^2P)4p-3p^5(^2P)3d$ and $3p^5(^2P)4p-3p^5(^2P)5p$. Similarly, for the level 5p, we calculated all the transitions $3p^5(^2P)5s-3p^5(^2P)5p$ and $3p^5(^2P)5p-3p^5(^2P)6s$. This is why we have this enormous number of transitions.

The 4s and 4p levels generate spectral lines that often appear in various applications. 4s-4p transition appears in the visible region, while the transition 3p-4s appears in the ultraviolet regions. The ten 4p-levels cannot directly decay to the 3p ground level due to parity selection rule, while they can effortlessly decay to the lower 4s-levels via sixteen allowed transitions due to jk -coupling. Seven of these transitions attend the two metastable 4s-levels. By all of the 4p-levels, the 3S_1 and 3P_2 decay easily to the metastable 3P_2 . Moreover, the 3D_3 decay basically only to 3P_2 . These transitions are responsible of the quasimetastable character of 3S_1 , 3P_2 and 3D_3 levels.

Table 2. The oscillator strength f_{ij} relative to $3p^6\ ^1S_0$ of neutral Argon atom Ar I, as obtained in the length $f_{ij}(l)$ and velocity $f_{ij}(v)$ forms of the electric dipole operator in Babushkin and Coulomb gauge, respectively.

Multiplet/Terms	g_i	g_k	Type	$E\ (\text{cm}^{-1})$	$f_{ij}(l)$	$f_{ij}(v)$
$3p^6\ ^1S_0$	1	0.0				
$3p^6\text{-}3p^5(^2P)4s$						
$^1S_0\text{-}^3P_1$	1	3	E ₁	93768.17	0.067075	0.051223
$^1S_0\text{-}^1P_1$	1	3	E ₁	95409.34	0.032639	0.018073
$3p^6\text{-}3p^5(^2P)3d$						
$^1S_0\text{-}^3P_1$	1	3	E ₁	112569.47	0.002744	0.002415
$^1S_0\text{-}^3D_1$	1	3	E ₁	114160.68	0.096092	0.084249
$^1S_0\text{-}^1P_1$	1	3	E ₁	115718.48	0.033697	0.029539
$3p^5(^2P)4s\text{-}3p^5(^2P)4p$						
$^3P_0\text{-}^3S_1$	1	3	E ₁	9793.91	0.04768	0.05367
$^3P_0\text{-}^3D_1$	1	3	E ₁	11509.82	0.09175	0.07424
$^3P_0\text{-}^1P_1$	1	3	E ₁	12573.81	0.73149	0.49598
$^3P_0\text{-}^3P_1$	1	3	E ₁	12924.40	0.40860	0.26401
$^3P_1\text{-}^3P_0$	3	1	E ₁	13302.17	0.04521	0.27477
$^3P_1\text{-}^1S_0$	3	1	E ₁	15132.92	0.003315	0.00164
$^1P_1\text{-}^3P_0$	3	1	E ₁	11661.00	0.002573	0.00195
$^1P_1\text{-}^1S_0$	3	1	E ₁	13491.75	0.45872	0.26997
$^3P_1\text{-}^3S_1$	3	3	E ₁	10628.50	0.23963	0.22915
$^3P_1\text{-}^3D_1$	3	3	E ₁	12344.40	0.93583	0.65949
$^3P_1\text{-}^1P_1$	3	3	E ₁	13408.40	0.00160	0.01009
$^3P_1\text{-}^3P_1$	3	3	E ₁	13758.98	0.05708	0.03220
$^1P_1\text{-}^3S_1$	3	3	E ₁	8987.32	0.01073	0.01411
$^1P_1\text{-}^3D_1$	3	3	E ₁	10703.23	0.04555	0.043005
$^1P_1\text{-}^1P_1$	3	3	E ₁	11767.22	0.05016	0.038725
$^1P_1\text{-}^3P_1$	3	3	E ₁	12117.81	0.66123	0.48433
$^3P_1\text{-}^3D_2$	3	5	E ₁	11865.78	1.38944	1.05754
$^3P_1\text{-}^3P_2$	3	5	E ₁	12458.80	0.18029	0.25905
$^3P_1\text{-}^1D_2$	3	5	E ₁	13534.39	0.44887	0.26450
$^1P_1\text{-}^3D_2$	3	5	E ₁	10224.61	0.12329	0.12599
$^1P_1\text{-}^3P_2$	3	5	E ₁	10817.63	0.37173	0.34258
$^1P_1\text{-}^1D_2$	3	5	E ₁	11893.22	1.48188	1.12798
$^3P_2\text{-}^3S_1$	5	3	E ₁	11234.47	0.83165	0.70641
$^3P_2\text{-}^3D_1$	5	3	E ₁	12950.38	0.18921	0.12007
$^3P_2\text{-}^1P_1$	5	3	E ₁	14014.37	0.02442	0.01322
$^3P_2\text{-}^3P_1$	5	3	E ₁	14364.96	0.17648	0.09169
$^3P_2\text{-}^3D_2$	5	5	E ₁	12471.76	0.52331	0.35889
$^3P_2\text{-}^3P_2$	5	5	E ₁	13064.78	1.51366	0.95401
$^3P_2\text{-}^1D_2$	5	5	E ₁	14140.36	0.18734	0.10057
$^3P_2\text{-}^3D_3$	5	7	E ₁	12323.22	2.95039	2.09289
$3p^5(^2P)4p\text{-}3p^5(^2P)3d$						
$^3P_0\text{-}^3P_1$	1	3	E ₁	5499.13	0.06973	0.13581
$^3P_0\text{-}^3D_1$	1	3	E ₁	7090.34	0.76498	0.89529
$^3P_0\text{-}^1P_1$	1	3	E ₁	8648.14	0.00687	0.00573
$^1S_0\text{-}^3P_1$	1	3	E ₁	3668.38	0.00125	0.00566
$^1S_0\text{-}^3D_1$	1	3	E ₁	5259.59	0.00783	0.01452

Table 2. Continued.

Multiplet/Terms	g_i	g_k	Type	E (cm ⁻¹)	$f_{ij}(l)$	$f_{ij}(v)$
¹ S ₀ - ¹ P ₁	1	3	E ₁	6817.39	0.81426	1.03674
³ S ₁ - ³ P ₀	3	1	E ₁	8043.87	0.42168	0.38531
³ D ₁ - ³ P ₀	3	1	E ₁	6327.97	0.06513	0.09719
¹ P ₁ - ³ P ₀	3	1	E ₁	5263.97	0.00645	0.01394
³ P ₀ - ³ P ₁	1	3	E ₁	4913.38	0.04274	1.04690
³ S ₁ - ³ P ₁	3	3	E ₁	8172.80	1.13148	1.0019
³ S ₁ - ³ D ₁	3	3	E ₁	9764.01	0.06973	0.1165
³ S ₁ - ¹ P ₁	3	3	E ₁	11321.81	1.536e ⁻⁴	7.987e ⁻⁵
³ D ₁ - ³ P ₁	3	3	E ₁	6456.90	0.02833	0.04121
³ D ₁ - ³ D ₁	3	3	E ₁	8048.10	0.70373	0.64202
³ D ₁ - ¹ P ₁	3	3	E ₁	9605.91	0.02215	0.01392
¹ P ₁ - ³ P ₁	3	3	E ₁	5392.90	0.02050	0.04229
¹ P ₁ - ³ D ₁	3	3	E ₁	6984.11	0.00721	0.00856
¹ P ₁ - ¹ P ₁	3	3	E ₁	8541.91	0.30487	0.24804
³ P ₁ - ³ P ₁	3	3	E ₁	5042.31	0.04600	0.10699
³ P ₁ - ³ D ₁	3	3	E ₁	6633.52	0.06039	0.08135
³ P ₁ - ¹ P ₁	3	3	E ₁	8191.32	0.43764	0.38387
³ S ₁ - ³ P ₂	3	5	E ₁	8432.40	1.41182	1.16824
³ S ₁ - ³ F ₂	3	5	E ₁	9011.22	2.262e ⁻⁴	2.381e ⁻⁴
³ S ₁ - ¹ D ₂	3	5	E ₁	10211.75	1.755e ⁻⁵	2.312e ⁻⁴
³ S ₁ - ³ D ₂	3	5	E ₁	10339.10	0.05114	0.02799
³ D ₁ - ³ P ₂	3	5	E ₁	6716.49	0.09403	0.12134
³ D ₁ - ³ F ₂	3	5	E ₁	7295.31	1.72901	1.91903
³ D ₁ - ¹ D ₂	3	5	E ₁	8495.85	0.03204	0.02651
³ D ₁ - ³ D ₂	3	5	E ₁	8623.20	0.03785	0.02952
¹ P ₁ - ³ P ₂	3	5	E ₁	5652.5	0.02936	0.05337
¹ P ₁ - ³ F ₂	3	5	E ₁	6231.32	1.416e ⁻⁴	1.895e ⁻⁴
¹ P ₁ - ¹ D ₂	3	5	E ₁	7431.85	2.34012	2.51000
¹ P ₁ - ³ D ₂	3	5	E ₁	7559.21	0.01001	0.01004
³ P ₁ - ³ P ₂	3	5	E ₁	5301.91	0.00867	0.01798
³ P ₁ - ³ F ₂	3	5	E ₁	5880.73	8.886e ⁻⁵	1.0138e ⁻⁴
³ P ₁ - ¹ D ₂	3	5	E ₁	7081.27	0.00203	0.00220
³ P ₁ - ³ D ₂	3	5	E ₁	7208.62	2.03287	2.29804
³ D ₂ - ³ P ₁	5	3	E ₁	6935.52	0.05642	0.07000
³ D ₂ - ³ D ₁	5	3	E ₁	8526.72	0.08219	0.06704
³ D ₂ - ¹ P ₁	5	3	E ₁	10084.52	0.00115	6.7622e ⁻⁴
³ P ₂ - ³ P ₁	5	3	E ₁	6342.50	0.22333	0.32647
³ P ₂ - ³ D ₁	5	3	E ₁	7933.70	0.00924	0.00868
³ P ₂ - ¹ P ₁	5	3	E ₁	9491.51	0.02464	0.01605
¹ D ₂ - ³ P ₁	5	3	E ₁	5266.91	0.03171	0.06733
¹ D ₂ - ³ D ₁	5	3	E ₁	6858.12	6.963e ⁻⁴	8.513e ⁻⁴
¹ D ₂ - ¹ P ₁	5	3	E ₁	8415.92	0.03120	0.02587
³ D ₂ - ³ P ₂	5	5	E ₁	7195.11	0.00821	0.00972
³ D ₂ - ³ F ₂	5	5	E ₁	7773.93	0.8678	0.86780
³ D ₂ - ¹ D ₂	5	5	E ₁	8974.47	0.0480	0.03551
³ D ₂ - ³ D ₂	5	5	E ₁	9101.82	0.0205	0.01469

Table 2. Continued.

Multiplet/Terms	g_i	g_k	Type	$E \text{ (cm}^{-1}\text{)}$	$f_{ij}(l)$	$f_{ij}(v)$
$^3P_2\text{-}^3P_2$	5	5	E ₁	6602.09	0.8442	1.13787
$^3P_2\text{-}^3F_2$	5	5	E ₁	7180.91	0.05608	0.06444
$^3P_2\text{-}^1D_2$	5	5	E ₁	8381.45	$3.524e^{-3}$	$2.910e^{-3}$
$^3P_2\text{-}^3D_2$	5	5	E ₁	8508.80	0.31702	0.25717
$^1D_2\text{-}^3P_2$	5	5	E ₁	5526.51	0.17771	0.34209
$^1D_2\text{-}^3F_2$	5	5	E ₁	6105.32	$1.127e^{-3}$	$1.868e^{-3}$
$^1D_2\text{-}^1D_2$	5	5	E ₁	7305.86	0.27724	0.30529
$^1D_2\text{-}^3D_2$	5	5	E ₁	7433.21	0.22465	0.23848
$^3D_2\text{-}^3F_3$	5	7	E ₁	7419.84	3.32815	3.58343
$^3D_2\text{-}^1F_3$	5	7	E ₁	7881.17	0.13714	0.12948
$^3D_2\text{-}^3D_3$	5	7	E ₁	9069.41	$8.195e^{-3}$	$5.656e^{-3}$
$^3P_2\text{-}^3F_3$	5	7	E ₁	6826.82	$1.957e^{-3}$	$2.686e^{-3}$
$^3P_2\text{-}^1F_3$	5	7	E ₁	7288.15	2.85489	3.15458
$^3P_2\text{-}^3D_3$	7	5	E ₁	8476.39	0.02611	0.02124
$^1D_2\text{-}^3F_3$	5	7	E ₁	5751.23	$8.149e^{-5}$	$9.762e^{-5}$
$^1D_2\text{-}^1F_3$	5	7	E ₁	6212.56	$5.962e^{-7}$	$9.716e^{-7}$
$^1D_2\text{-}^3D_3$	5	7	E ₁	7400.80	3.67883	3.94317
$^3D_3\text{-}^3P_2$	7	5	E ₁	7343.65	0.10948	0.11916
$^3D_3\text{-}^3F_2$	7	5	E ₁	7922.47	0.05326	0.04980
$^3D_3\text{-}^1D_2$	7	5	E ₁	9123.01	$5.667e^{-3}$	$0.3997e^{-3}$
$^3D_2\text{-}^3D_2$	5	5	E ₁	9250.36	0.01900	0.01303
$^3D_3\text{-}^3F_3$	7	7	E ₁	7568.38	0.45173	0.46299
$^3D_3\text{-}^1F_3$	7	7	E ₁	8029.70	0.81270	0.73978
$^3D_3\text{-}^3D_3$	7	7	E ₁	9217.94	0.08499	0.05871
$^3D_3\text{-}^3F_4$	7	9	E ₁	7395.52	4.85257	5.20721
$3p^5(^2P) 4p\text{-}3p^5(^2P)5s$						
$^3S_1\text{-}^3P_0$	3	1	E ₁	10025.2	0.03712	0.01818
$^3D_1\text{-}^3P_0$	3	1	E ₁	8309.29	0.03712	0.01818
$^1P_1\text{-}^3P_0$	3	1	E ₁	7245.30	0.31634	0.30049
$^3P_1\text{-}^3P_0$	3	1	E ₁	6894.71	0.16569	0.17173
$^3P_1\text{-}^3P_1$	3	3	E ₁	6115.99	0.13032	0.17242
$^1S_0\text{-}^1P_1$	1	3	E ₁	4285.23	0.01965	0.05276
$^3P_0\text{-}^3P_1$	1	3	E ₁	7483.86	1.48241	1.32890
$^1S_0\text{-}^3P_1$	1	3	E ₁	5653.11	0.11918	0.18653
$^3S_1\text{-}^1P_1$	3	3	E ₁	8789.66	0.11482	0.07083
$^3D_1\text{-}^1P_1$	3	3	E ₁	7073.75	0.42068	0.41755
$^1P_1\text{-}^1P_1$	3	3	E ₁	6009.76	0.03036	0.04160
$^3P_1\text{-}^1P_1$	3	3	E ₁	5659.17	$1.832e^{-3}$	$2.760e^{-3}$
$^3S_1\text{-}^3P_1$	3	3	E ₁	10157.53	0.05440	0.02623
$^3D_1\text{-}^3P_0$	3	1	E ₁	8441.63	$7.213e^{-3}$	$5.144e^{-3}$
$^1P_1\text{-}^3P_1$	3	3	E ₁	7377.64	0.19828	0.18290
$^3P_1\text{-}^3P_1$	3	3	E ₁	7027.05	0.30905	0.31058
$^3D_2\text{-}^1P_1$	5	3	E ₁	7552.37	0.72988	0.63880
$^3P_2\text{-}^1P_1$	5	3	E ₁	6959.35	0.21048	0.21406
$^1D_2\text{-}^1P_1$	5	3	E ₁	5883.76	0.01072	0.01510
$^3D_2\text{-}^3P_1$	5	3	E ₁	8920.25	$1.647e^{-3}$	$0.997e^{-3}$

Table 2. Continued.

Multiplet/Terms	g_i	g_k	Type	$E \text{ (cm}^{-1}\text{)}$	$f_{ij}(l)$	$f_{ij}(v)$
$^3P_2\text{-}^3P_1$	5	3	E ₁	8327.23	0.09705	0.06915
$^1D_2\text{-}^3P_1$	5	3	E ₁	7251.64	0.85310	0.80236
$^3S_1\text{-}^3P_2$	3	5	E ₁	8580.38	0.47678	0.32288
$^3D_1\text{-}^3P_2$	3	5	E ₁	6864.47	0.07429	0.07959
$^1P_1\text{-}^3P_2$	3	5	E ₁	5800.48	7.478e^{-3}	0.01123
$^3P_1\text{-}^3P_2$	3	5	E ₁	5449.89	0.05026	0.08436
$^3D_2\text{-}^3P_2$	5	5	E ₁	7343.09	0.22919	0.21375
$^3P_2\text{-}^3P_2$	5	5	E ₁	6750.07	0.59075	0.64260
$^1D_2\text{-}^3P_2$	5	5	E ₁	5674.48	0.05656	0.08739
$^3D_3\text{-}^3P_2$	7	5	E ₁	7491.63	1.35817	1.19662
$3p^6\text{-}3p^5(^2P)5s$						
$^1S_0\text{-}^1P_1$	1	3	E ₁	113186.32	0.02956	0.027065
$^1S_0\text{-}^3P_1$	1	3	E ₁	114554.20	0.02356	0.021615
$3p^6\text{-}3p^5(^2P)4d$						
$^1S_0\text{-}^3P_1$	1	3	E ₁	118472.16	1.786e^{-3}	1.575e^{-3}
$^1S_0\text{-}^3D_1$	1	3	E ₁	119443.99	0.07579	0.06665
$^1S_0\text{-}^1P_1$	1	3	E ₁	120884.20	0.17226	0.15144
$3p^5(^2P)5s\text{-}3p^5(^2P)5p$						
$^3P_0\text{-}^3S_1$	1	3	E ₁	1535.08	8.278e^{-3}	0.03609
$^3P_0\text{-}^1P_1$	1	3	E ₁	2002.92	3.966e^{-3}	0.01007
$^3P_0\text{-}^3D_1$	1	3	E ₁	3274.71	0.93772	0.89066
$^3P_0\text{-}^3P_1$	1	3	E ₁	3338.70	0.61077	0.56265
$^1P_1\text{-}^3P_0$	3	1	E ₁	3665.04	0.57374	0.43759
$^1P_1\text{-}^1S_0$	3	1	E ₁	5026.31	0.01114	4.557e^{-3}
$^3P_1\text{-}^3P_0$	3	1	E ₁	2297.16	5.130e^{-3}	9.798e^{-3}
$^3P_1\text{-}^1S_0$	3	1	E ₁	3658.43	0.57404	0.43591
$^1P_1\text{-}^3S_1$	3	3	E ₁	2770.62	0.21397	0.28721
$^1P_1\text{-}^1P_1$	3	3	E ₁	3238.46	1.29230	1.25914
$^1P_1\text{-}^3D_1$	3	3	E ₁	4510.26	4.716e^{-5}	2.692e^{-5}
$^1P_1\text{-}^3P_1$	3	3	E ₁	4574.25	8.730e^{-4}	4.138e^{-4}
$^3P_1\text{-}^3P_1$	3	3	E ₁	1402.75	7.097e^{-3}	0.07097
$^3P_1\text{-}^1P_1$	3	3	E ₁	1870.58	3.069e^{-3}	9.026e^{-3}
$^3P_1\text{-}^3D_1$	3	3	E ₁	3142.38	0.59948	0.61565
$^3P_1\text{-}^3P_1$	3	3	E ₁	3206.37	0.89641	0.89144
$^1P_1\text{-}^3D_2$	3	5	E ₁	3079.08	1.85322	1.98996
$^1P_1\text{-}^3P_2$	3	5	E ₁	3261.18	0.60153	0.58148
$^1P_1\text{-}^1D_2$	3	5	E ₁	4564.16	0.03846	0.01898
$^3P_1\text{-}^3D_2$	3	5	E ₁	1711.20	4.107e^{-3}	0.01419
$^3P_1\text{-}^3P_2$	3	5	E ₁	1893.31	0.01920	0.05498
$^3P_1\text{-}^1D_2$	3	5	E ₁	3196.28	2.49713	2.50462
$^3P_2\text{-}^3S_1$	5	1	E ₁	2979.90	1.15953	1.33051
$^3P_0\text{-}^1P_1$	1	3	E ₁	3447.74	0.02559	0.02175
$^3P_2\text{-}^3D_1$	5	3	E ₁	4719.54	1.034e^{-3}	4.671e^{-3}
$^3P_2\text{-}^3P_1$	5	3	E ₁	4783.53	0.06540	0.02912
$^3P_2\text{-}^3D_2$	5	5	E ₁	3288.36	0.62914	0.58993
$^3P_2\text{-}^3P_2$	5	5	E ₁	3470.46	2.07997	1.76643



Table 2. Continued.

Multiplet/Terms	g_i	g_k	Type	$E \text{ (cm}^{-1}\text{)}$	$f_{ij}(l)$	$f_{ij}(v)$
$^3\text{P}_2\text{-}^1\text{D}_2$	5	5	E ₁	4773.44	0.01960	8.773e ⁻³
$^3\text{P}_2\text{-}^3\text{D}_3$	5	7	E ₁	3232.35	3.59213	3.52309
$3\text{p}^5(^2\text{P})5\text{p}\text{-}3\text{p}^5(^2\text{P})6\text{s}$						
$^3\text{P}_0\text{-}^1\text{P}_1$	1	3	E ₁	2096.77	0.25590	0.37771
$^3\text{P}_0\text{-}^3\text{P}_1$	1	3	E ₁	3502.50	0.01587	8.514e ⁻³
$^1\text{S}_0\text{-}^1\text{P}_1$	1	3	E ₁	735.50	3.385e ⁻³	0.04049
$^1\text{S}_0\text{-}^3\text{P}_1$	1	3	E ₁	2141.22	0.25862	0.37020
$^3\text{S}_1\text{-}^3\text{P}_0$	3	1	E ₁	4348.81	0.01879	6.422e ⁻³
$^1\text{P}_1\text{-}^3\text{P}_0$	3	1	E ₁	3880.98	6.076e ⁻³	2.638e ⁻³
$^3\text{D}_1\text{-}^3\text{P}_0$	3	1	E ₁	2609.18	0.59050	0.56745
$^3\text{P}_1\text{-}^3\text{P}_0$	3	1	E ₁	2545.19	0.37265	0.37218
$^3\text{S}_1\text{-}^1\text{P}_1$	3	3	E ₁	2991.19	0.17809	0.12814
$^3\text{S}_1\text{-}^3\text{P}_1$	3	3	E ₁	4396.91	0.02899	9.784e ⁻³
$^1\text{P}_1\text{-}^1\text{P}_1$	3	3	E ₁	2523.35	0.80308	0.82145
$^1\text{P}_1\text{-}^3\text{P}_1$	3	3	E ₁	3929.07	9.209e ⁻⁵	4.159e ⁻⁵
$^3\text{D}_1\text{-}^1\text{P}_1$	3	3	E ₁	1251.55	1.232e ⁻⁵	5.087e ⁻³
$^3\text{D}_1\text{-}^3\text{P}_1$	3	3	E ₁	2657.28	0.39534	0.36852
$^3\text{P}_1\text{-}^1\text{P}_1$	3	3	E ₁	1187.56	5.940e ⁻⁴	2.680e ⁻³
$^3\text{P}_1\text{-}^3\text{P}_1$	3	3	E ₁	2593.29	0.57541	0.55694
$^3\text{S}_1\text{-}^3\text{P}_2$	3	5	E ₁	2902.99	0.89326	0.69309
$^1\text{P}_1\text{-}^3\text{P}_2$	3	5	E ₁	2435.16	0.14105	0.15739
$^3\text{D}_1\text{-}^3\text{P}_2$	3	5	E ₁	1163.36	1.976e ⁻⁴	9.720e ⁻⁴
$^3\text{P}_1\text{-}^3\text{P}_2$	3	5	E ₁	1099.37	0.01188	0.06430
$^3\text{D}_2\text{-}^1\text{P}_1$	5	3	E ₁	2682.73	1.28003	1.16439
$^3\text{D}_2\text{-}^3\text{P}_1$	5	3	E ₁	4088.45	2.021e ⁻⁴	7.531e ⁻⁵
$^3\text{P}_2\text{-}^1\text{P}_1$	5	3	E ₁	2500.63	0.38057	0.39314
$^3\text{P}_2\text{-}^3\text{P}_1$	5	3	E ₁	3906.35	0.01491	6.333e ⁻³
$^1\text{D}_2\text{-}^1\text{P}_1$	5	3	E ₁	1197.65	7.162e ⁻⁴	3.187e ⁻³
$^1\text{D}_2\text{-}^3\text{P}_1$	5	3	E ₁	2603.37	1.63853	1.56864
$^3\text{D}_2\text{-}^3\text{P}_2$	5	5	E ₁	2594.53	0.38888	0.38088
$^3\text{P}_2\text{-}^3\text{P}_2$	5	5	E ₁	2412.43	1.15020	1.28552
$^1\text{D}_2\text{-}^3\text{P}_2$	5	5	E ₁	1109.45	3.607e ⁻³	0.01914
$^3\text{D}_3\text{-}^3\text{P}_2$	7	5	E ₁	2650.54	2.34989	2.17024
$3\text{p}^6\text{-}3\text{p}^5(^2\text{P})6\text{s}$						
$^1\text{S}_0\text{-}^1\text{P}_1$	1	3	E ₁	118948.13	0.012660	0.01233
$^1\text{S}_0\text{-}^3\text{P}_1$	1	3	E ₁	120353.86	0.01338	0.01463

Furthermore, the two transitory 4s-levels $^3\text{P}_1[3/2]_1$ and $^1\text{P}_1[1/2]_1$ are strongly mixing by spin-orbit interaction as well as the $4\text{p}^3\text{D}_1[3/2]_1$, $4\text{p}^1\text{P}_1[3/2]_1$, $4\text{p}^3\text{P}_0[1/2]_0$ and $4\text{p}^1\text{S}_0[1/2]_1$ levels.

As a gauge of convergence and precision of our results, we illustrated in tables 3, 4, 5, 6, 7 and 8 the comparison between our results and those either measured experimentally or predicted theoretically for the $3\text{p}^6\text{ }^1\text{S}_0\text{-}3\text{p}^5n\text{s}^3\text{P}_1$ and the $3\text{p}^6\text{ }^1\text{S}_0\text{-}3\text{p}^5n\text{s}^1\text{P}_1$ transition array with $n = 4, 5$ and 6. Various experimental techniques have been used to measure the optical oscillator strength as shown in these tables and the corresponding references therein. As an example of these experimental techniques, Chan and co-workers [71] used dipole (e, e) spectrometer to measure the absolute photoabsorption oscillator strength for argon in the energy range 16–500 eV. The self-absorption of the resonance radiation method was employed to measure the optical oscillator strengths of noble gas resonance lines by Lintenberg [23], Gibson and Risely [24]. The absolute oscillator strengths for the discrete and autoionizing resonances

Table 3. Comparison between experimental and theoretical values of the oscillator strength for the $3p^6\ ^1S_0$ - $3p^54s\ ^3P_1$ transition array.

Reference	Experimental		Theoretical	
	f_{ij}	$f_{ij}(l)$	f_{ij}	$f_{ij}(v)$
Present Work		0.067075		0.051223
Copley <i>et al.</i> [17]	0.076 ± 0.008			
Vallee <i>et al.</i> [18]	0.051 ± 0.007			
Westerveld <i>et al.</i> [19]	0.063 ± 0.005			
Chornay <i>et al.</i> [20]	0.065 ± 0.005			
Wu <i>et al.</i> [22]	0.0676 ± 0.0040			
Ligtenberg <i>et al.</i> [23]	0.0616 ± 0.0021			
Gibson <i>et al.</i> [24]	0.0580 ± 0.0017			
Lee <i>et al.</i> [28]	0.07 ± 0.007		0.08	
Lee [29]			0.059	
Savukov <i>et al.</i> [32]			0.0629	
Irimia <i>et al.</i> [33]		0.0619		0.057
NIST [38]			0.0609	
Froese Fischer <i>et al.</i> [67]			0.05455	
Aymar <i>et al.</i> [68]		0.071		0.065
Weis <i>et al.</i> [69]				0.061
R.S. Knox [70]				0.052
Chan <i>et al.</i> [71]	0.0662 ± 0.0033			
Tsurubuchi <i>et al.</i> [72]	0.057 ± 0.003			
Li <i>et al.</i> [73]	$0.058^{+0.005}_{-0.008}$			
Avgoustoglou <i>et al.</i> [74]		0.0672		0.0670
Hahn <i>et al.</i> [75]	0.077 ± 0.004			
Federman <i>et al.</i> [76]	0.064 ± 0.003			
Lang <i>et al.</i> [77]	0.083 ± 0.027			
Geiger [78]	0.047 ± 0.009			
Zatsarinny <i>et al.</i> [79]		0.0678		0.0602

Table 4. Comparison between experimental and theoretical values of the oscillator strength for the $3p^6\ ^1S_0$ - $3p^54s\ ^1P_1$ transition array.

Reference	Experimental		Theoretical	
	f_{ij}	$f_{ij}(l)$	f_{ij}	$f_{ij}(v)$
Present Work		0.032639		0.018073
Westerveld <i>et al.</i> [19]	0.02			
Wu <i>et al.</i> [22]	0.015			
Lee [29]			0.030	
NIST [38]			0.025	
Chan <i>et al.</i> [71]	0.013			
Lawrence [80]	0.021			

were reported by Wu and co-workers by means of electron-energy-loss spectroscopy [22]. Westerveld and coworkers [19] measured experimentally the oscillator strengths using self-absorption of resonance radiation for fourteen resonance transitions of neon and argon.

For a direct comparison with experimental data, all the results are reported in terms of absolute oscillator strength, whereas the theoretical results are given in terms of Babushkin and Coulomb gauges or simply reported as oscillator strength only. In the latter, it is not clear how the oscillator strengths are calculated. From our point of view, it is the average of the oscillator strengths in Babushkin and Coulomb gauge. In spite of these comparisons, a noticeable disparity is observed in the experimental-experimental data, theoretical-theoretical data as well as experimental-

Table 5. Comparison between experimental and theoretical values of the oscillator strength for the $3p^6\ ^1S_0\text{-}3p^55s\ ^3P_1$ transition array.

Reference	Experimental		Theoretical	
	f_{ij}	$f_{ij}(l)$	f_{ij}	$f_{ij}(v)$
Present Work		0.02706		0.02995
Westerveld <i>et al.</i> [19]	0.025 ± 0.005			
Wu <i>et al.</i> [22]	0.0241			
Lee <i>et al.</i> [28]			0.045	
Lee [29]	0.028 ± 0.003		0.034	
Irimia <i>et al.</i> [33]			0.0331	
Aymar [61]		0.028		0.030
W.F. Chan <i>et al.</i> [71]	0.0264 ± 0.0033			
Geiger [78]	0.032 ± 0.009			
Zatsarinny <i>et al.</i> [79]		0.0340		0.0301
Lawrence [80]	0.0268 ± 0.002			
Wiese <i>et al.</i> [81]	0.028			

Table 6. Comparison between experimental and theoretical values of the oscillator strength for the $3p^6\ ^1S_0\text{-}3p^55s\ ^1P_1$ transition array.

Reference	Experimental		Theoretical	
	f_{ij}	$f_{ij}(l)$	f_{ij}	$f_{ij}(v)$
Present Work		0.021615		0.02356
Westerveld <i>et al.</i> [19]	0.0106 ± 0.005			
Wu <i>et al.</i> [22]	0.0122			
Lee <i>et al.</i> [28]	0.0124 ± 0.004		0.039	
Lee [29]			0.025	
NIST [38]			0.0119	
Aymar [61]		0.0217		0.0227
Chan <i>et al.</i> [71]	0.0126 ± 0.0033			
Geiger [78]	0.0108 ± 0.009			
Zatsarinny <i>et al.</i> [79]		0.0180		0.0160
Lawrence [80]	0.0119 ± 0.002			
Wiese <i>et al.</i> [81]	0.013			

Table 7. Comparison between experimental and theoretical values of the oscillator strength for the $3p^6\ ^1S_0\text{-}3p^56s\ ^3P_1$ transition array.

Reference	Experimental		Theoretical	
	f_{ij}	$f_{ij}(l)$	f_{ij}	$f_{ij}(v)$
Present Work		0.012660		0.01233
Wu <i>et al.</i> [22]	0.0129			
Lee [29]	0.0094 ± 0.007		0.023	
Chan <i>et al.</i> [71]	0.0144 ± 0.0033			
Zatsarinny <i>et al.</i> [79]		0.0135		0.0119

theoretical data. It is obvious that the slight dispersion between experimental data gives us a test of the quality of our theoretical results. Also, in each experimental result, one theoretical treatment seems to be better than the others to represent the experimental results, but it is not always the same one at all time. Nevertheless, our results either for the length form or velocity form of the oscillator strengths seems to be in good agreement with the experimental results as well as the theoretical ones.

Table 8. Comparison between experimental and theoretical values of the oscillator strength for the $3p^6\ ^1S_0$ - $3p^56s\ ^1P_1$ transition array.

Reference	Experimental	Theoretical	
	f_{ij}	$f_{ij}(l)$	$f_{ij}(v)$
Present Work		0.01338	0.01463
Wu <i>et al.</i> [22]	0.0223		
Lee <i>et al.</i> [28]	0.022 ± 0.004		0.013
Chan <i>et al.</i> [71]	0.0221 ± 0.0033		
Zatsarinny <i>et al.</i> [79]		0.0337	0.0296

Unfortunately, these are the only levels that we found in the literature. The lack of experimental results of oscillator strengths and transition rates for the levels in question is due to the excitation energies. These levels are autoionized and their absorption spectrums are increasingly overlapping and difficult to resolve as the excitation energy approaches the limits of ionization. **The lack of experimental results for both oscillator strengths and transition rates suggests that more appropriate theoretical models are necessary for spectrum calculations that included the highly excited states as well as the ground states.**

6.3 Computation of the radiative transition rates

We computed the radiative transition rates A_{fi} , (in s^{-1}) of the intermediate Rydberg series $ns\ [k]_J$ ($4 \leq n \leq 6$), $nd\ [k]_J$ ($3 \leq n \leq 4$), $np\ [k]_J$ ($4 \leq n \leq 5$) relative to the ground state $3p^6\ ^1S_0$ of neutral argon atom. These transition rates were calculated in the length $A_{fi}(l)$ and velocity $A_{fi}(v)$ of the electric dipole operator forms in Babushkin and Coulomb gauge, respectively. The obtained results are tabulated in table 9. Certain indication of the accuracy of theoretical radiative transition values is afforded by the agreement of the theoretical lifetimes with the corresponding ones obtained either theoretically or experimentally. The inverse of the lifetime of a level f is given by

$$\frac{1}{\tau_f} = \sum_i A_{fi}, \quad (76)$$

where the sum runs over all the lower energy levels connected to level f by a radiative transition. In table 10 we compare our theoretical results of the lifetimes with those obtained by various authors. The experimental results are obtained by various authors using different experimental techniques. In general, our values agree with both the experimental ones and the other theoretical results.

6.4 Computation of Landé g -factor and the hyperfine structure constants

The relativistic Landé g -factor is calculated for the odd series under prediction relative to the ground level $3p^6\ ^1S_0$. The obtained results are tabulated in table 11. As there is no theoretical data of the Landé g -factor, many experimental ones have been left without comparison with the theoretical ones. However, our results are compared to those obtained, experimentally by Salah [14], Chenevier and Moskowitz [36], Abu Safia [41, 92] and Moore [93–95] or theoretically by Aymar and Schweighofer [37, 90], Abu Safia [41] as well as NIST [38] as shown in table 12. The comparison shows an acceptable agreement between these results. The Landé g -factor for the levels $3d[1/2]_1$, $4d[1/2]_1$, $5d[1/2]_1$, $10d[1/2]_1$ and $11d[1/2]_1$ are not measured by Salah, while he measured the Landé g -factor for the levels $6d[1/2]_1$, $7d[1/2]_1$, $8d[1/2]_1$ and $9d[1/2]_1$ by using optogalvanique atomic laser spectroscopy. The uncertainty of his results is estimated to be 0.02. On the other side, Chenevier and Moskowitz [9] measured experimentally the Landé g -factor for the levels $3d[1/2]_1$ using optically detected magnetic resonance technique. The theoretical value of this level is determined by Aymar and Schweighofe [7]. Abu Safia measured and theoretically calculated the Landé g -factor for the level $4p[1/2]_1$ [92], $4p'[1/2]_1$, $4p'[3/2]_2$, $4p'[3/2]_1$, $4p[3/2]_2$, $4p[3/2]_1$ and $4p[5/2]_2$ [41]. In his measurements, Abu Safia used Laser optical pumping and magnetic resonance technique.

The magnetic dipole A_{MD} and the electric quadrupole B_{EQ} hyperfine constants are calculated by GRASP. The obtained results are tabulated in table 11.

It is worth mentioning that the theoretical results of Landé g -factor for the $ns\ [k]_J$ ($4 \leq n \leq 6$), $nd\ [k]_J$ ($3 \leq n \leq 4$), $np\ [k]_J$ ($4 \leq n \leq 5$) and the hyperfine structure constants are reported here for the first time.

Table 9. Radiative transition rates A_{fi} , (in s^{-1}) relative to $3p^6\ ^1\text{S}_0$ of neutral Argon atom Ar I, as obtained in the length $A_{if}(l)$ and velocity $A_{if}(v)$ forms of the electric dipole operator in Babushkin and Coulomb gauge, respectively.

Multiplet/Terms	g_i	g_k	Type	$E\ (\text{cm}^{-1})$	$A_{if}(l)$	$A_{if}(v)$
$3p^6\ ^1\text{S}_0$	1	0.0				
$3p^6\text{-}3p^5(^2\text{P})4s$						
$^1\text{S}_0\text{-}^3\text{P}_1$	1	3	E ₁	93768.17	$1.32233\ \text{e}^8$	$1.05510\ \text{e}^8$
$^1\text{S}_0\text{-}^1\text{P}_1$	1	3	E ₁	95409.34	$4.82507\ \text{e}^8$	$3.67276\ \text{e}^8$
$3p^6\text{-}3p^5(^2\text{P})3d$						
$^1\text{S}_0\text{-}^3\text{P}_1$	1	3	E ₁	112569.47	$14.46626\ \text{e}^6$	$13.61012\ \text{e}^6$
$^1\text{S}_0\text{-}^3\text{D}_1$	1	3	E ₁	114160.68	$2.78447\ \text{e}^8$	$2.44127\ \text{e}^8$
$^1\text{S}_0\text{-}^1\text{P}_1$	1	3	E ₁	115718.48	$3.00328\ \text{e}^8$	$2.93158\ \text{e}^8$
$3p^5(^2\text{P})4s\text{-}3p^5(^2\text{P})4p$						
$^3\text{P}_0\text{-}^3\text{S}_1$	1	3	E ₁	9793.91	$1.01689\ \text{e}^6$	$1.14477\ \text{e}^6$
$^3\text{P}_0\text{-}^3\text{D}_1$	1	3	E ₁	11509.82	$2.70247\ \text{e}^6$	$2.18674\ \text{e}^6$
$^3\text{P}_0\text{-}^1\text{P}_1$	1	3	E ₁	12573.81	$2.57135\ \text{e}^7$	$1.74348\ \text{e}^7$
$^3\text{P}_0\text{-}^3\text{P}_1$	1	3	E ₁	12924.40	$1.51756\ \text{e}^7$	$9.80753\ \text{e}^7$
$^3\text{P}_1\text{-}^3\text{P}_0$	3	1	E ₁	13302.17	$5.33627\ \text{e}^7$	$3.24311\ \text{e}^7$
$^3\text{P}_1\text{-}^1\text{S}_0$	3	1	E ₁	15132.92	$5.06401\ \text{e}^5$	$2.46531\ \text{e}^5$
$^1\text{P}_1\text{-}^3\text{P}_0$	3	1	E ₁	11661.00	$2.33374\ \text{e}^5$	$1.77261\ \text{e}^5$
$^1\text{P}_1\text{-}^1\text{S}_0$	3	1	E ₁	13491.75	$5.56965\ \text{e}^7$	$3.27821\ \text{e}^7$
$^3\text{P}_1\text{-}^3\text{S}_1$	3	3	E ₁	10628.50	$6.01897\ \text{e}^6$	$5.75555\ \text{e}^6$
$^3\text{P}_1\text{-}^3\text{D}_1$	3	3	E ₁	12344.40	$3.17075\ \text{e}^7$	$2.23447\ \text{e}^7$
$^3\text{P}_1\text{-}^1\text{P}_1$	3	3	E ₁	13408.40	$6.41079\ \text{e}^4$	$4.03647\ \text{e}^4$
$^3\text{P}_1\text{-}^3\text{P}_1$	3	3	E ₁	13758.98	$2.40291\ \text{e}^6$	$1.3577\ \text{e}^6$
$^1\text{P}_1\text{-}^3\text{S}_1$	3	3	E ₁	8987.32	$1.92659\ \text{e}^5$	$2.53408\ \text{e}^5$
$^1\text{P}_1\text{-}^3\text{D}_1$	3	3	E ₁	10703.23	$1.16031\ \text{e}^6$	$1.09540\ \text{e}^6$
$^1\text{P}_1\text{-}^1\text{P}_1$	3	3	E ₁	11767.22	$1.54437\ \text{e}^7$	$1.19224\ \text{e}^7$
$^1\text{P}_1\text{-}^3\text{P}_1$	3	3	E ₁	12117.81	$2.15887\ \text{e}^7$	$1.58129\ \text{e}^7$
$^3\text{P}_1\text{-}^3\text{D}_2$	3	5	E ₁	11865.78	$2.60979\ \text{e}^7$	$1.98637\ \text{e}^7$
$^3\text{P}_1\text{-}^3\text{P}_2$	3	5	E ₁	12458.80	$5.36436\ \text{e}^6$	$3.73340\ \text{e}^6$
$^3\text{P}_1\text{-}^1\text{D}_2$	3	5	E ₁	13534.39	$1.09693\ \text{e}^7$	$6.46362\ \text{e}^6$
$^1\text{P}_1\text{-}^3\text{D}_2$	3	5	E ₁	10224.61	$1.71953\ \text{e}^6$	$1.75720\ \text{e}^6$
$^1\text{P}_1\text{-}^3\text{P}_2$	3	5	E ₁	10817.63	$5.80321\ \text{e}^6$	$5.34817\ \text{e}^6$
$^1\text{P}_1\text{-}^1\text{D}_2$	3	5	E ₁	11893.22	$2.79630\ \text{e}^7$	$2.12849\ \text{e}^7$
$^3\text{P}_2\text{-}^3\text{S}_1$	5	3	E ₁	11234.47	$2.33382\ \text{e}^7$	$1.98238\ \text{e}^7$
$^3\text{P}_2\text{-}^3\text{D}_1$	5	3	E ₁	12950.38	$7.05558\ \text{e}^6$	$4.47756\ \text{e}^6$
$^3\text{P}_2\text{-}^1\text{P}_1$	5	3	E ₁	14014.37	$1.06624\ \text{e}^5$	$5.77482\ \text{e}^5$
$^3\text{P}_2\text{-}^3\text{P}_1$	5	3	E ₁	14364.96	$8.09720\ \text{e}^6$	$4.20686\ \text{e}^6$
$^3\text{P}_2\text{-}^3\text{D}_2$	5	5	E ₁	12471.76	$1.08591\ \text{e}^7$	$7.44723\ \text{e}^6$
$^3\text{P}_2\text{-}^3\text{P}_2$	5	5	E ₁	13064.78	$3.44671\ \text{e}^7$	$2.17236\ \text{e}^7$
$^3\text{P}_2\text{-}^1\text{D}_2$	5	5	E ₁	14140.36	$4.99727\ \text{e}^6$	$2.68283\ \text{e}^6$
$^3\text{P}_2\text{-}^3\text{D}_3$	5	7	E ₁	12323.22	$4.26946\ \text{e}^7$	$3.02857\ \text{e}^6$
$3p^5(^2\text{P})4p\text{-}3p^5(^2\text{P})3d$						
$^3\text{P}_0\text{-}^3\text{P}_1$	1	3	E ₁	5499.13	$4.68817\ \text{e}^5$	$9.13139\ \text{e}^5$
$^3\text{P}_0\text{-}^3\text{D}_1$	1	3	E ₁	7090.34	$8.55082\ \text{e}^6$	$10.0074\ \text{e}^6$
$^3\text{P}_0\text{-}^1\text{P}_1$	1	3	E ₁	8648.14	$1.14267\ \text{e}^5$	$0.95533\ \text{e}^5$
$^1\text{S}_0\text{-}^3\text{P}_1$	1	3	E ₁	3668.38	$0.37443\ \text{e}^4$	$1.69576\ \text{e}^4$
$^1\text{S}_0\text{-}^3\text{D}_1$	1	3	E ₁	5259.59	$4.35921\ \text{e}^4$	$8.92638\ \text{e}^4$

Table 9. Continued.

Multiplet/Terms	g_i	g_k	Type	E (cm ⁻¹)	$A_{if}(l)$	$A_{if}(v)$
¹ S ₀ - ¹ P ₁	1	3	E ₁	6817.39	0.84143 e ⁷	1.07134 e ⁷
³ S ₁ - ³ P ₀	3	1	E ₁	8043.87	1.81994 e ⁷	1.66298 e ⁷
³ D ₁ - ³ P ₀	3	1	E ₁	6327.97	1.73973 e ⁶	2.59607 e ⁶
¹ P ₁ - ³ P ₀	3	1	E ₁	5263.97	1.19373 e ⁵	2.57629 e ⁵
³ P ₀ - ³ P ₁	1	3	E ₁	4913.38	0.68837 e ⁶	1.68581 e ⁶
³ S ₁ - ³ P ₁	3	3	E ₁	8172.80	1.68038 e ⁷	1.48541 e ⁷
³ S ₁ - ³ D ₁	3	13	E ₁	9764.01	1.2532 e ⁵	0.729269 e ⁵
³ S ₁ - ¹ P ₁	3	3	E ₁	11321.81	4.38018 e ³	2.27643 e ³
³ D ₁ - ³ P ₁	3	3	E ₁	6456.90	2.62610 e ⁵	3.82024 e ⁵
³ D ₁ - ³ D ₁	3	3	E ₁	8048.10	10.1348 e ⁶	9.24613 e ⁶
³ D ₁ - ¹ P ₁	3	3	E ₁	9605.91	4.54582 e ⁵	2.85557 e ⁵
¹ P ₁ - ³ P ₁	3	3	E ₁	5392.90	1.32587 e ⁵	2.73526 e ⁵
¹ P ₁ - ³ D ₁	3	3	E ₁	6984.11	7.8254 e ⁴	9.31221 e ⁴
¹ P ₁ - ¹ P ₁	3	3	E ₁	8541.91	4.94604 e ⁶	4.02408 e ⁶
³ P ₁ - ³ P ₁	3	3	E ₁	5042.31	2.60094 e ⁵	6.04840 e ⁵
³ P ₁ - ³ D ₁	3	3	E ₁	6633.52	5.90940 e ⁵	7.95947 e ⁵
³ P ₁ - ¹ P ₁	3	3	E ₁	8191.32	6.52900 e ⁶	5.72689 e ⁶
³ S ₁ - ³ P ₂	3	5	E ₁	8432.40	1.33923 e ⁷	1.10817 e ⁷
³ S ₁ - ³ F ₂	3	5	E ₁	9011.22	2.45040 e ³	2.57995 e ³
³ S ₁ - ¹ D ₂	3	5	E ₁	10211.75	2.44181 e ²	3.21750 e ²
³ S ₁ - ³ D ₂	3	5	E ₁	10339.10	7.29295 e ⁵	3.99179 e ⁵
³ D ₁ - ³ P ₂	3	5	E ₁	6716.49	5.65894 e ⁵	7.30267 e ⁵
³ D ₁ - ³ F ₂	3	5	E ₁	7295.31	1.22760 e ⁷	1.36252 e ⁷
³ D ₁ - ¹ D ₂	3	5	E ₁	8495.85	3.08568 e ⁵	2.55295 e ⁵
³ D ₁ - ³ D ₂	3	5	E ₁	8623.20	3.75524 e ⁵	2.92883 e ⁵
¹ P ₁ - ³ P ₂	3	5	E ₁	5652.5	1.25185 e ⁵	2.27510 e ⁵
¹ P ₁ - ³ F ₂	3	5	E ₁	6231.32	7.33928 e ²	9.81768 e ²
¹ P ₁ - ¹ D ₂	3	5	E ₁	7431.85	1.72427 e ⁷	1.84944 e ⁷
¹ P ₁ - ³ D ₂	3	5	E ₁	7559.21	7.63383 e ⁴	7.65434 e ⁴
³ P ₁ - ³ P ₂	3	5	E ₁	5301.91	3.25172 e ⁴	6.74276 e ⁴
³ P ₁ - ³ F ₂	3	5	E ₁	5880.73	4.10005 e ²	4.67764 e ²
³ P ₁ - ¹ D ₂	3	5	E ₁	7081.27	1.36332 e ⁴	1.47690 e ⁴
³ P ₁ - ³ D ₂	3	5	E ₁	7208.62	1.40924 e ⁷	1.59307 e ⁷
³ D ₂ - ³ P ₁	5	3	E ₁	6935.52	6.03752 e ⁵	7.48666 e ⁵
³ D ₂ - ³ D ₁	5	3	E ₁	8526.72	1.32878 e ⁶	1.08387 e ⁶
³ D ₂ - ³ P ₁	5	3	E ₁	10084.52	2.62256 e ⁴	1.52905 e ⁴
³ P ₂ - ³ P ₁	5	3	E ₁	6342.50	1.99752 e ⁶	2.92003 e ⁶
³ P ₂ - ³ D ₁	5	3	E ₁	7933.70	1.29351 e ⁵	1.21527 e ⁵
³ P ₂ - ¹ P ₁	5	3	E ₁	9491.51	4.93706 e ⁵	3.21491 e ⁵
¹ D ₂ - ³ P ₁	5	3	E ₁	5266.91	1.95639 e ⁵	4.15572 e ⁵
¹ D ₂ - ³ D ₁	5	3	E ₁	6858.12	7.28169 e ³	8.90305 e ³
¹ D ₂ - ¹ P ₁	5	3	E ₁	8415.92	4.91352 e ⁵	4.07440 e ⁵
³ D ₂ - ³ P ₂	5	5	E ₁	7195.11	5.87181 e ⁴	6.71814 e ⁴
³ D ₂ - ³ F ₂	5	5	E ₁	7773.93	6.99642 e ⁶	6.86653 e ⁶
³ D ₂ - ¹ D ₂	5	5	E ₁	8974.47	5.16223 e ⁵	3.81629 e ⁵
³ D ₂ - ³ D ₂	5	5	E ₁	9101.82	2.26565 e ⁵	1.62402 e ⁵

Table 9. Continued.

Multiplet/Terms	g_i	g_k	Type	E (cm ⁻¹)	$A_{if}(l)$	$A_{if}(v)$
³ P ₂ - ³ P ₂	5	5	E ₁	6602.09	4.90892 e ⁶	6.61648 e ⁶
³ P ₂ - ³ F ₂	5	5	E ₁	7180.91	3.85789 e ⁵	4.43321 e ⁵
³ P ₂ - ¹ D ₂	5	5	E ₁	8381.45	3.30317 e ⁴	2.72751 e ⁴
³ P ₂ - ³ D ₂	5	5	E ₁	8508.80	3.06195 e ⁶	2.48392 e ⁶
¹ D ₂ - ³ P ₂	5	5	E ₁	5526.51	0.72408 e ⁶	1.39385 e ⁶
¹ D ₂ - ³ F ₂	5	5	E ₁	6105.32	5.60551 e ³	9.29063 e ³
¹ D ₂ - ¹ D ₂	5	5	E ₁	7305.86	1.97416 e ⁶	2.17386 e ⁶
¹ D ₂ - ³ D ₂	5	5	E ₁	7433.21	1.65585 e ⁶	1.75786 e ⁶
³ D ₂ - ³ F ₃	5	7	E ₁	7419.84	1.74597 e ⁷	1.87989 e ⁷
³ D ₂ - ¹ F ₃	5	7	E ₁	7881.17	8.11730 e ⁵	7.66380 e ⁵
³ D ₂ - ³ D ₃	5	7	E ₁	9069.41	6.42350 e ⁴	4.43317 e ⁴
³ P ₂ - ³ F ₃	5	7	E ₁	6826.82	0.86938 e ⁴	1.19291 e ⁴
³ P ₂ - ¹ F ₃	5	7	E ₁	7288.15	1.44500 e ⁷	1.59669 e ⁷
³ D ₃ - ³ P ₂	7	5	E ₁	8476.39	1.78765 e ⁵	1.45446 e ⁵
¹ D ₂ - ³ F ₃	5	7	E ₁	5751.23	2.56869 e ²	3.07713 e ²
¹ D ₂ - ¹ F ₃	5	7	E ₁	6212.56	2.19289	3.57369
¹ D ₂ - ³ D ₃	5	7	E ₁	7400.80	1.92004 e ⁷	2.05801 e ⁷
³ D ₃ - ³ P ₂	7	5	E ₁	7343.65	7.87663 e ⁵	8.57308 e ⁵
³ D ₃ - ³ F ₂	7	5	E ₁	7922.47	4.45759 e ⁵	4.16988 e ⁵
³ D ₃ - ¹ D ₂	7	5	E ₁	9123.01	6.29225 e ⁴	4.43889 e ⁴
³ D ₂ - ³ D ₂	5	5	E ₁	9250.36	2.16927 e ⁵	1.48804 e ⁵
³ D ₃ - ³ F ₃	7	7	E ₁	7568.38	2.46564 e ⁶	2.52712 e ⁶
³ D ₃ - ¹ F ₃	7	7	E ₁	8029.70	4.99318 e ⁶	4.54517 e ⁶
³ D ₃ - ³ D ₃	7	7	E ₁	9217.94	6.88191 e ⁵	4.75402 e ⁵
³ D ₃ - ³ F ₄	7	9	E ₁	7395.52	1.96702 e ⁷	2.11077 e ⁷
3p⁵(²P)4p-3p⁵(²P)5s						
³ S ₁ - ³ P ₀	3	1	E ₁	10025.2	2.48839 e ⁶	1.21928 e ⁶
³ D ₁ - ³ P ₀	3	1	E ₁	8309.29	2.28932 e ⁶	1.65348 e ⁶
¹ P ₁ - ³ P ₀	3	1	E ₁	7245.30	1.10768 e ⁶	1.05220 e ⁶
³ P ₁ - ³ P ₀	3	1	E ₁	6894.71	5.25382 e ⁶	5.44533 e ⁶
³ P ₁ - ³ P ₁	3	3	E ₁	6115.99	1.08386 e ⁶	1.43403 e ⁶
¹ S ₀ - ¹ P ₁	1	3	E ₁	4285.23	0.82363 e ⁵	2.15427 e ⁵
³ P ₀ - ³ P ₁	1	3	E ₁	7483.86	4.19654 e ⁵	3.76198 e ⁵
³ S ₀ - ³ P ₁	1	3	E ₁	5653.11	8.46853 e ⁵	13.2543 e ⁵
³ S ₁ - ¹ P ₁	3	3	E ₁	8789.66	1.91500 e ⁶	1.21675 e ⁶
³ D ₁ - ¹ P ₁	3	3	E ₁	7073.75	4.68035 e ⁶	4.64556 e ⁶
¹ P ₁ - ¹ P ₁	3	3	E ₁	6009.76	2.43827 e ⁵	3.34077 e ⁵
³ P ₁ - ¹ P ₁	3	3	E ₁	5659.17	1.30513 e ⁴	196562 e ⁴
³ S ₁ - ³ P ₁	3	3	E ₁	10157.53	1.24809 e ⁶	0.60188 e ⁶
³ D ₁ - ³ P ₀	3	1	E ₁	8441.63	1.14291 e ⁵	0.81504 e ⁵
¹ P ₁ - ³ P ₁	3	3	E ₁	7377.64	2.39966 e ⁶	2.21335 e ⁶
³ P ₁ - ³ P ₁	3	3	E ₁	7027.05	3.39309 e ⁶	3.40994 e ⁶
³ D ₂ - ¹ P ₁	5	3	E ₁	7552.37	9.25639 e ⁶	8.10131 e ⁶
³ P ₂ - ¹ P ₁	5	3	E ₁	6959.35	2.26662 e ⁶	2.30519 e ⁶
¹ D ₂ - ¹ P ₁	5	3	E ₁	5883.76	0.82554 e ⁵	1.16268 e ⁵
³ D ₂ - ³ P ₁	5	3	E ₁	8920.25	2.91443 e ⁴	1.76441 e ⁴

Table 9. Continued.

Multiplet/Terms	g_i	g_k	Type	E (cm ⁻¹)	$A_{if}(l)$	$A_{if}(v)$
³ P ₂ - ³ P ₁	5	3	E ₁	8327.23	1.49638 e ⁶	1.06628 e ⁶
¹ D ₁ - ³ P ₁	3	3	E ₁	7251.64	9.97461 e ⁶	9.38140 e ⁶
³ S ₁ - ³ P ₂	3	5	E ₁	8580.38	4.68283 e ⁶	3.17125 e ⁶
³ D ₁ - ³ P ₂	3	5	E ₁	6864.47	4.67040 e ⁵	5.00364 e ⁵
¹ P ₁ - ³ P ₂	3	5	E ₁	5800.48	3.35683 e ⁴	5.04158 e ⁴
³ P ₁ - ³ P ₂	3	5	E ₁	5449.89	1.99149 e ⁵	3.34290 e ⁵
³ D ₂ - ³ P ₂	5	5	E ₁	7343.09	1.64871 e ⁶	1.53758 e ⁶
³ S ₂ - ³ P ₂	5	5	E ₁	6750.07	3.59084 e ⁶	3.90599 e ⁶
¹ D ₂ - ³ P ₂	5	5	E ₁	5674.48	2.42985 e ⁵	3.75409 e ⁵
³ D ₃ - ³ P ₂	7	5	E ₁	7491.63	1.01690 e ⁶	8.95942 e ⁶
3p ⁶ -3p ⁵ (² P)5s						
¹ S ₀ - ¹ P ₁	1	3	E ₁	113186.32	8.41104 e ⁷	7.69889 e ⁷
¹ S ₀ - ³ P ₁	1	3	E ₁	114554.20	5.71431 e ⁷	5.52486 e ⁷
3p ⁶ -3p ⁵ (² P)4d						
¹ S ₀ - ³ P ₁	1	3	E ₁	118472.16	5.5786 e ⁶	4.91708 e ⁶
¹ S ₀ - ³ D ₁	1	3	E ₁	119443.99	2.4043 e ⁸	2.11436 e ⁸
¹ S ₀ - ¹ P ₁	1	3	E ₁	120884.20	5.5970 e ⁸	4.92047 e ⁸
3p ⁵ (² P)5s-3p ⁵ (² P)5p						
³ P ₀ - ³ S ₁	1	3	E ₁	1535.08	0.43373 e ⁴	1.89142 e ⁴
³ P ₀ - ¹ P ₁	1	3	E ₁	2002.92	3.53826 e ³	8.89479 e ³
³ P ₀ - ³ D ₁	1	3	E ₁	3274.71	2.23585 e ⁶	2.12365 e ⁶
³ P ₀ - ³ P ₁	1	3	E ₁	3338.70	1.51376 e ⁶	1.39449 e ⁶
¹ P ₁ - ³ P ₀	3	1	E ₁	3665.04	5.14066 e ⁶	3.92075 e ⁶
¹ P ₁ - ¹ S ₀	3	1	E ₁	5026.31	1.87819 e ⁵	0.76809 e ⁵
³ P ₁ - ³ P ₀	3	1	E ₁	2297.16	1.80598 e ⁴	3.44901 e ⁴
³ P ₁ - ¹ S ₀	3	1	E ₁	3658.43	5.12480 e ⁶	3.89161 e ⁶
¹ P ₁ - ³ S ₁	3	3	E ₁	2770.62	3.65200 e ⁵	4.90215 e ⁵
¹ P ₁ - ¹ P ₁	3	3	E ₁	3238.46	3.01343 e ⁶	2.93610 e ⁶
¹ P ₁ - ³ D ₁	3	3	E ₁	4510.26	2.13346 e ²	1.21790 e ²
¹ P ₁ - ³ P ₁	3	3	E ₁	4574.25	4.406159 e ³	1.92546 e ³
³ P ₁ - ³ P ₁	3	3	E ₁	1402.75	0.10528 e ⁴	1.60636 e ⁴
³ P ₁ - ¹ P ₁	3	3	E ₁	1870.58	2.38839 e ³	7.02259 e ³
³ P ₁ - ³ D ₁	3	3	E ₁	3142.38	1.31618 e ⁶	1.35116 e ⁶
³ P ₁ - ³ P ₁	3	3	E ₁	3206.37	2.04907 e ⁶	2.03770 e ⁶
¹ P ₁ - ³ D ₂	3	5	E ₁	3079.08	2.34391 e ⁶	2.51686 e ⁶
¹ P ₁ - ³ P ₂	3	5	E ₁	3261.18	8.53458 e ⁵	8.25014 e ⁵
¹ P ₁ - ¹ D ₂	3	5	E ₁	4564.16	1.06900 e ⁵	0.52771 e ⁵
³ P ₁ - ³ D ₂	3	5	E ₁	1711.20	1.60449 e ³	5.54569 e ³
³ P ₁ - ³ P ₂	3	5	E ₁	1893.31	0.91182 e ⁴	2.62927 e ⁴
³ P ₁ - ¹ D ₂	3	5	E ₁	3196.28	3.40333 e ⁶	3.41353 e ⁶
³ P ₂ - ³ S ₁	5	3	E ₁	2979.90	2.28933 e ⁶	2.62690 e ⁶
³ P ₀ - ¹ P ₁	1	3	E ₁	3447.74	6.76514 e ⁵	5.74879 e ⁵
³ P ₂ - ³ D ₁	5	3	E ₁	4719.54	5.12365 e ³	2.31333 e ³
³ P ₂ - ³ P ₁	5	3	E ₁	4783.53	3.32740 e ⁵	1.48161 e ⁵
³ P ₂ - ³ D ₂	5	5	E ₁	3288.36	8.50038 e ⁵	9.07577 e ⁵
³ P ₂ - ³ P ₂	5	5	E ₁	3470.46	3.34198 e ⁶	2.83820 e ⁶



Table 9. Continued.

Multiplet/Terms	g_i	g_k	Type	E (cm ⁻¹)	$A_{if}(l)$	$A_{if}(v)$
$^3P_2-^1D_2$	5	5	E ₁	4773.44	5.96016 e ⁴	2.66677 e ⁴
$^3P_2-^3D_3$	5	7	E ₁	3232.35	3.57629 e ⁶	3.50756 e ⁶
$3p^5(^2P)5p-3p^5(^2P)6s$						
$^3P_0-^1P_1$	1	3	E ₁	2096.77	2.50511 e ⁵	3.69220 e ⁵
$^3P_0-^3P_1$	1	3	E ₁	3502.50	4.32899 e ⁴	2.32244 e ⁴
$^1S_0-^1P_1$	1	3	E ₁	735.50	0.407167 e ³	4.87045 e ³
$^1S_0-^3P_1$	1	3	E ₁	2141.22	2.63647 e ⁵	3.77385 e ⁵
$^3S_1-^3P_0$	3	1	E ₁	4348.81	2.37039 e ⁵	0.810135 e ⁵
$^1P_1-^3P_0$	3	1	E ₁	3880.98	6.10455 e ⁴	2.65096 e ⁴
$^3D_1-^3P_0$	3	1	E ₁	2609.18	2.68148 e ⁶	2.57678 e ⁶
$^3P_1-^3P_0$	3	1	E ₁	2545.19	1.61021 e ⁶	1.60820 e ⁶
$^3S_1-^1P_1$	3	3	E ₁	2991.19	3.54287 e ⁵	2.54913 e ⁵
$^3S_1-^3P_1$	3	3	E ₁	4396.91	1.24645 e ⁵	0.420598 e
$^1P_1-^1P_1$	3	3	E ₁	2523.35	1.13694 e ⁶	1.16295 e ⁶
$^1P_1-^3P_1$	3	3	E ₁	3929.07	3.16110 e ²	1.42762 e ²
$^3D_1-^1P_1$	3	3	E ₁	1251.55	0.42915 e ³	1.77181 e ³
$^3D_1-^3P_1$	3	3	E ₁	2657.28	6.20687 e ⁵	5.78581 e ⁵
$^3P_1-^1P_1$	3	3	E ₁	1187.56	1.86282 e ²	8.40447 e ²
$^3P_1-^3P_1$	3	3	E ₁	2593.29	8.60412 e ⁵	8.32784 e ⁵
$^3S_1-^3P_2$	3	5	E ₁	2902.99	1.00426 e ⁶	0.77921 e ⁶
$^1P_1-^3P_2$	3	5	E ₁	2435.16	1.11587 e ⁵	1.24515 e ⁵
$^3D_1-^3P_2$	3	5	E ₁	1163.36	0.35678 e ²	1.75496 e ²
$^3P_1-^3P_2$	3	5	E ₁	1099.37	19.1635 e ⁴	1.03684 e ⁴
$^3D_2-^1P_1$	5	3	E ₁	2682.73	2.04830 e ⁶	1.86326 e ⁶
$^3D_2-^3P_1$	5	3	E ₁	4088.45	7.51257 e ²	2.79924 e ²
$^3P_2-^1P_1$	5	3	E ₁	2500.63	5.29120 e ⁵	5.56598 e ⁵
$^3P_2-^3P_1$	5	3	E ₁	3906.35	5.06032 e ⁴	2.14888 e ⁴
$^1D_2-^1P_1$	5	3	E ₁	1197.65	0.22842 e ³	1.01654 e ³
$^1D_2-^3P_1$	5	3	E ₁	2603.37	2.6916 e ⁶	2.36383 e ⁶
$^3D_2-^3P_2$	5	5	E ₁	2594.53	3.49228 e ⁵	3.42044 e ⁵
$^3P_2-^3P_2$	5	5	E ₁	2412.43	8.93008 e ⁵	9.98067 e ⁵
$^1D_2-^3P_2$	5	5	E ₁	1109.45	0.59229 e ³	3.14395 e ³
$^3D_3-^3P_2$	7	5	E ₁	2650.54	2.20237 e ⁶	20.3399 e ⁶
$3p^6-3p^5(^2P)6s$						
$^1S_0-^1P_1$	1	3	E ₁	118948.13	3.54410 e ⁷	3.88193 e ⁷
$^1S_0-^3P_1$	1	3	E ₁	120353.86	2.35604 e ⁷	0.01463 e ⁷

7 Conclusion

In the framework of the MCDHF approach including Breit interaction, quantum electrodynamics (QED) effects and self-energy correction, we tentatively predicted the atomic data of the intermediate Rydberg series $ns[k]_J$ ($4 \leq n \leq 6$), $nd[k]_J$ ($3 \leq n \leq 4$), $np[k]_J$ ($4 \leq n \leq 5$) relative to the ground state $3p^6\ ^1S_0$ of neutral argon atom. The predicted atomic data are the energy levels, the oscillator strengths f_{ij} , the radiative transition rates A_{ij} , the Landé g -factor, the magnetic dipole moment and the electric quadrupole hyperfine constants.

The accuracy of the computed energy levels is afforded by the agreement with the corresponding ones measured experimentally [16] or calculated theoretically [67]. In general, our theoretical values agree much better with the experimental results than the other theoretical ones.

Table 10. Comparison between experimental and theoretical values of the life times (ns) in the length T_l and velocity T_v forms of some levels in neutral argon atom.

Level	Reference	Theoretical values				Experimental values
		T_l	$\tau_{radiative}$	T_v	$T_{radiative} = \left(\frac{T_l + T_v}{2}\right)$	$\tau_{radiative}$
4s[3/2] ₁	Present work	7.5624		9.9452	8.7538	
	Irimia <i>et al.</i> [33]				9.65	
	Zatsarinny <i>et al.</i> [79]	7.55		8.551	8.05	
	Wiese <i>et al.</i> [69]		9.4 ± 1.7			
	Knox [70]	9.84		10.44	10.14	
	Chronay <i>et al.</i> [20]					7.9 ± 0.6
	Lawrence [80]					8.6 ± 0.4
	Hahn <i>et al.</i> [75]					6.61 ± 0.4
	Burnham <i>et al.</i> [82]					9.5 ± 1.1
	Irwin <i>et al.</i> [83]					6.2 ± 0.2
	Lewis [84]					8.12 ± 0.5
	Kurucz [91]		4.405			
4s'[1/2] ₁	Present work	2.0725		2.7227	2.3976	
	Irimia <i>et al.</i> [33]				1.86	
	Zatsarinny <i>et al.</i> [79]	1.93		2.17	2.05	
	Lawrence [80]					2.15 ± 0.2
	Kurucz [91]		1.059			
3d[1/2] ₁	Present work	69.126		73.474	71.3	
	Irimia <i>et al.</i> [33]				68.0	
	Zatsarinny <i>et al.</i> [79]	71.4		76.0	73.7	
	Kurucz [91]			71.492		
4p[1/2] ₁	Present work	40.664		47.345	44.004	
	Irimia <i>et al.</i> [33]				39.0	
	Zatsarinny <i>et al.</i> [79]	36.3		34.5	35.4	
	Kurucz [91]		32.68			
	Wiese <i>et al.</i> [85]					40.5
	Chang <i>et al.</i> [86]					42.2
	Shumaker <i>et al.</i> [89]					38.2
4p[5/2] ₃	Present work	30.586		33.018	31.802	
	Irimia <i>et al.</i> [33]					27.9
	Zatsarinny <i>et al.</i> [79]	27.1		26.3		
	Kurucz [91]		24.631			
	Wiese <i>et al.</i> [85]					30.2
	Chang <i>et al.</i> [86]					30.1
	Lilly [87]			28.7		
	Shumaker <i>et al.</i> [89]					29.9

Table 10. Continued.

Level	Reference	Theoretical values				Experimental values
		T_l	$\tau_{radiative}$	T_v	$T_{radiative} = \left(\frac{T_l + T_v}{2}\right)$	$\tau_{radiative}$
4p[5/2] ₂	Present work	25.855		32.161	29.008	
	Irimia <i>et al.</i> [33]				29.0	
	Zatsarinny <i>et al.</i> [79]	28.9		28.1	28.5	
	Kurucz [91]		26.525			
	Lilly [87]				28.7	
	Chang <i>et al.</i> [86]					29.8
	Wiese <i>et al.</i> [85]					30.6
4p[3/2] ₁	Present work	23.649		33.359	28.504	
	Irimia <i>et al.</i> [33]				27.8	
	Zatsarinny <i>et al.</i> [79]	27.6		26.8	27.2	
	Kurucz [91]		25.253			
	Wiese [85]					30.2
	Lilly [87]				28.7	
	Chang <i>et al.</i> [86]					29.3
	Musiol <i>et al.</i> [88]					30.6
4p[3/2] ₂	Present work	21.913		32.462	27.187	
	Irimia <i>et al.</i> [33]				26.4	
	Zatsarinny <i>et al.</i> [79]	26.7		26.2	26.45	
	Kurucz [91]		24.096			
	Wiese [85]					29.4
	Lilly [87]				27.0	
	Chang <i>et al.</i> [86]					27.0
	Shumaker <i>et al.</i> [89]					29.9
4p[1/2] _o	Present work	17.792		28.370	23.08	
	Irimia <i>et al.</i> [33]				23.1	
	Zatsarinny <i>et al.</i> [79]	23.0		23.1	26.45	
	Kurucz [91]		21.598			
	Wiese [85]					24.4
	Lilly [87]				22.8	
	Musiol <i>et al.</i> [88]					24.8
	Shumaker <i>et al.</i> [89]					25.5
4p'[3/2] ₁	Present work	23.649		33.361	28.505	
	Irimia <i>et al.</i> [33]				27.6	
	Zatsarinny <i>et al.</i> [79]	28.1		27.2	27.65	
	Kurucz [91]		25.445			
	Wiese [85]					29.3
	Chang <i>et al.</i> [86]					33.2
	Lilly [87]				29.0	
	Musiol <i>et al.</i> [88]					30.6
	Shumaker <i>et al.</i> [89]					31.3

Table 10. Continued.

Level	Reference	Theoretical values				Experimental values
		T_l	$\tau_{radiative}$	T_v	$T_{radiative} = \left(\frac{T_l + T_v}{2}\right)$	$\tau_{radiative}$
$4p[3/2]_2$	Present work	22.763		32.856	27.814	
	Irimia <i>et al.</i> [33]				29.0	
	Zatsarinny <i>et al.</i> [79]	27.1		26.5	27.65	
	Kurucz [91]		24.691			
	Wiesse [85]					29.0
	Chang <i>et al.</i> [86]					27.6
	Lilly [87]				27.6	
	Shumaker <i>et al.</i> [89]					29.4
$4p[1/2]_1$	Present work	23.649		33.361	28.505	
	Irimia <i>et al.</i> [33]				26.2	
	Zatsarinny <i>et al.</i> [79]	25.6		25.0	25.3	
	Kurucz [91]		23.753			
	Wiesse [85]					28.3
	Chang <i>et al.</i> [86]					26.5
	Lilly [87]					25.6
	Musiol <i>et al.</i> [88]					25.3
$4p[1/2]_0$	Present work	18.658		30.666	24.662	
	Irimia <i>et al.</i> [33]				21.6	
	Zatsarinny <i>et al.</i> [79]	20.5		20.8	20.65	
	Kurucz [91]		21.413			
	Wiesse [85]					21.7
	Lilly [87]				22.6	
	Musiol <i>et al.</i> [88]					21.5
	Shumaker <i>et al.</i> [89]					23.0
$5s[1/2]_1$	Present work	17.499		18.100	17.799	
	Aymar <i>et al.</i> [90]				17.5	
	Lawrence [80]					17.5 ± 2
	Kurucz [91]		6.452			
$5s[3/2]_1$	Present work	11.889		12.988	12.43	
	Irimia <i>et al.</i> [33]				8.31	
	Zatsarinny <i>et al.</i> [79]	8.30		9.15	8.725	
	Kurucz [91]		3.472			
	Aymar <i>et al.</i> [90]				8.1	
	Lawrence [80]					10.1 ± 0.4

Table 11. Relativistic Landé g -factor and hyperfine structure constants of levels belonging to the intermediate Rydberg series $ns[k]_J$ ($4 \leq n \leq 6$), $nd[k]_J$ ($3 \leq n \leq 4$), $np[k]_J$ ($4 \leq n \leq 5$) relative to $3p^6 \ ^1S_0$ of neutral argon atom.

Designation	LS-Term	g_J	A_{MD} [MHz]	B_{QE} [MHz]
4s				
4s[3/2] ₁	3P_1	1.39273	29.550878	−21.182298
4s[1/2] ₁	1P_1	1.10809	664.66859	79.058037
4s[3/2] ₂	3P_2	1.50102	234.13922	115.751478
3d				
3d[1/2] ₁	3P_1	1.45930	−14.28108	−18.5228
3d[3/2] ₁	3D_1	0.62701	−62.7054	−21.5343
3d'[3/2] ₁	1P_1	0.91375	−285.78616	5.101737
3d[3/2] ₂	3P_2	1.43138	−19.992024	−30.688450
3d'[5/2] ₂	1D_2	0.86691	139.067991	43.054776
3d[5/2]	3F_2	0.86622	248.983049	−56.995603
3d'[3/2] ₂	3D_2	1.16943	−126.271182	−39.218693
3d[7/2] ₃	3F_3	1.06252	120.247747	111.975386
3d'[5/2] ₃	3D_3	1.19238	84.089509	19.603216
3d[5/2] ₃	1F_3	1.16253	187.907030	−44.746827
3d[7/2] ₄	3F_4	1.25050	89.436088	116.132228
4p				
4p[1/2] ₁	3S_1	1.96908	91.13018	0.61842
4p[3/2] ₁	3D_1	0.84054	290.436316	45.20373
4p'[3/2] ₁	1P_1	0.81899	459.664707	−33.41966
4p'[1/2] ₁	3P_1	1.37331	−105.654780	−1.00173
4p[5/2] ₂	3D_2	1.10797	180.211002	96.101919
4p[3/2] ₂	3P_2	1.30928	151.667509	60.012428
4p'[3/2] ₂	1D_2	1.25066	282.65472	−42.81507
4p[3/2] ₁	1P_1	1.33401	121.08902	114.0075
5s				
5s[3/2] ₁	3P_1	1.27611	648.57874	20.38691
5s[1/2] ₁	1P_1	1.2241	274.02373	37.78161
5s[3/2] ₂	3P_2	1.50110	193.06380	116.33706
4d				
4d[1/2] ₁	3P_1	1.42157	13.77128	−24.18941
4d[3/2] ₁	3D_1	0.68429	−82.365668	−14.72842
4d'[3/2] ₁	1P_1	0.894215	−295.16113	3.96328
4d[3/2] ₂	3P_2	1.38159	8.090169	−47.91414
4d'[5/2] ₂	1D_2	0.92867	116.18692	14.25617
4d[5/2] ₂	3F_2	0.80407	270.20704	−27.64573
4d'[3/2] ₂	3D_2	1.21879	−153.70831	−22.02512
4d[7/2] ₃	3F_3	1.06637	119.58714	115.19775
4d'[5/2] ₃	3D_3	1.22083	73.99501	−10.30738
4d[5/2] ₃	1F_3	1.13023	198.5736	−17.84424
4d[7/2] ₄	3F_4	1.25051	89.12305	116.27927
5p				
5p[1/2] ₁	3S_1	1.88818	160.58430	1.32375
5p[3/2] ₁	3D_1	1.01017	225.89522	19.13373
5p'[3/2] ₁	1P_1	0.64331	537.78810	−7.77090

Table 11. Continued.

Designation	LS-Term	g_J	A_{MD} [MHz]	B_{QE} [MHz]
5p'[1/2] ₁	³ P ₁	1.46027	−197.66830	−1.12122
5p[5/2] ₂	³ D ₂	1.09694	175.37322	104.28145
5p[3/2] ₂	³ P ₂	1.38951	128.30251	21.192711
5p'[3/2] ₂	¹ D ₂	1.181474	303.32165	−9.820547
5p[3/2] ₁	¹ P ₁	1.334015	119.60193	115.65362
6s				
6s[3/2] ₁	³ P ₁	1.31208	628.31137	7.72797
6s[1/2] ₁	¹ P ₁	1.18874	286.05165	50.49840
6s[3/2] ₂	³ P ₂	1.50102	184.28850	116.4527

Table 12. Comparison of Relativistic Landé g -factor of levels belonging to the intermediate Rydberg series $ns[k]_J$ ($4 \leq n \leq 6$), $nd[k]_J$ ($3 \leq n \leq 4$), $np[k]_J$ ($4 \leq n \leq 5$) with other experimental and theoretical results.

Designation	This work	Experimental work				Theoretical work			
		[41]	[92]	[93–95]	[36]	[38]	[41]	[37]	[90]
4s[3/2] ₁ ³ P ₁	1.39273					1.404			
4s[1/2] ₁ ¹ P ₁	1.10809					1.102			
4s[3/2] ₂ ³ P ₂	1.50102					1.506			
3d[1/2] ₁ ³ P ₁	1.45930				1.4885(20)			1.481	
3d[3/2] ₁ ³ D ₁	0.62701							0.713	
3d'[3/2] ₁ ¹ P ₁	0.91375								
3d[3/2] ₂ ³ P ₂	1.43138							1.471	
3d'[5/2] ₂ ¹ D ₂	0.86691							0.928	
3d[5/2] ₂ ³ F ₂	0.86622							0.805	
3d'[3/2] ₂ ³ D ₂	1.1694							1.133	
4p[1/2] ₁ ³ S ₁	1.96908		1.974220(49)	1.985					1.974
4p[3/2] ₁ ³ D ₁	0.84054	0.82950(25)		0.838		0.838	0.823		0.822
4p'[3/2] ₁ ¹ P ₁	0.81899	0.81709(24)		0.819		0.819	0.806		0.837
4p'[1/2] ₁ ³ P ₁	1.37331	1.38003(40)		1.38		1.380	1.389		1.366
4p[5/2] ₂ ³ D ₂	1.10797	1.11221(33)		1.112		1.112	1.115		1.115
4p[3/2] ₂ ³ P ₂	1.30928	1.30130(39)		1.305		1.305	1.299		1.299
4p'[3/2] ₂ ¹ D ₂	1.25066	1.25384(37)		1.260		1.26	1.253		1.260
5s[3/2] ₁ ³ P ₁	1.27611								1.28
5s[1/2] ₁ ¹ P ₁	1.2241								1.218
5s[3/2] ₂ ³ P ₂	1.50110								1.496
4d[1/2] ₁ ³ P ₁	1.42157					1.467			
4d[3/2] ₂ ³ P ₂	1.38159					1.437			
4d[3/2] ₁ ¹ D ₃	0.68429					0.768			
4d[7/2] ₄ ³ F ₄	1.25051					1.255			
4d[7/2] ₃ ³ F ₃	1.06637					1.077			
4d[5/2] ₂ ³ F ₂	0.80407					0.908			
6s[3/2] ₁ ³ P ₁	1.31208					1.271			
6s[1/2] ₁ ¹ P ₁	1.18874					1.184			
6s[3/2] ₂ ³ P ₂	1.50102					1.500			

Based on the accuracy of our results, we computed the oscillator strengths f_{ij} , the radiative transition rates A_{ij} , the Landé g -factor and the magnetic dipole A_{MD} and the electric quadrupole B_{EQ} hyperfine constants for the levels under consideration. The accuracy of the obtained data of the oscillator strengths and the radiative transition rates has been assessed through comparisons with both the experimental and theoretical radiative lifetime measurements and predictions, respectively. An overall excellent agreement has been observed. Moreover, for a direct comparison of our theoretical radiative lifetime with the experimental data, it is evident that the theoretical radiative lifetime is the mean value of the lifetimes obtained in length and velocity gauges. Consequently, this excellent agreement, relieved doubt about the discrepancies of the relativistic oscillator strengths in length and velocity gauges, and provides an idea of the accuracy of the wave functions used in the calculations. On the other hand, the gauge invariance of the results is itself a necessity but not an adequate and sufficient condition to deduce conclusions about the reliability convergence of the results in the relativistic framework.

The obtained values of the Landé g -factor are compared with the available experimental and theoretical ones. A fairly excellent agreement between our results and the experimental and theoretical results is observed. This excellent agreement is a factor that assures the accuracy of our predicted data concerning the magnetic dipole A_{MD} and the electric quadrupole B_{EQ} hyperfine constants for the levels under consideration. Unfortunately, we were unable to compare our predicted results of the hyperfine constants with any other experimental or theoretical results because the required results are missing. However, our accurate values of Landé g -factor and hyperfine structure constants are of great interest for atomic spectroscopy researchers. They provide a precise interpretation of level-crossing experiment [39–41] as well as a sensitive test for wave functions [90, 96, 97].

Finally, we chose for our predictions those spectra which had not been sufficiently analyzed, or whose analysis showed essential disparities with the theoretical works or the analysis of closely related spectra. Our new, fairly complete, set of precise and reliable theoretical results could be of great interest for both experimental and theoretical research on neutral argon atoms. These data might be useful for astronomer in the identification of unknown lines encountered in the solar and stellar spectra. Moreover, these data should stimulate further theoretical work in the atomic spectroscopy field.

References

1. Xin Xu *et al.*, J. Phys. B **49**, 064010 (2016).
2. I.D. Petrov *et al.*, J. Phys. B **36**, 119 (2003).
3. A. Dasgupta *et al.*, Phys. Rev. A **61**, 012703 (1999).
4. J.W. Coburn, M. Chen, J. Appl. Phys. **51**, 3134 (1980).
5. S.A. Levshakov *et al.*, Astron. Astrophys. **373**, 836 (2001).
6. S. Lopez *et al.*, Astron. Astrophys. **385**, 778 (2002).
7. I. Santiago, M.D. Calzada, Appl. Spectrosc. **61**, 725 (2007).
8. M. Akel *et al.*, J. Fusion Energy **32**, 350 (2013).
9. J. Kuba *et al.*, J. Opt. Soc. Am. B **20**, 609 (2003).
10. L. Masoudnia, Phys. Plasmas **23**, 043108 (2016).
11. A.A. Lutman *et al.*, Phys. Rev. Lett. **110**, 134801 (2013).
12. K. Yoshino, J. Opt. Soc. Am. **60**, 1220 (1970).
13. G. Racah, Phys. Rev. **62**, 438 (1942).
14. W. Salah, Nucl. Instrum. Methods B **196**, 25 (2002).
15. M. Pellarin *et al.*, J. Phys. B **21**, 3833 (1988).
16. L. Minnhagen, J. Opt. Soc. Am. **63**, 1185 (1973).
17. G.H. Copley, D.M. Camm, J. Quant. Spectrosc. Radiat. Transf. **14**, 899 (1974).
18. O. Vallee *et al.*, J. Quant. Spectrosc. Radiat. Transf. **18**, 327 (1977).
19. W. Westerveld *et al.*, J. Quant. Spectrosc. Radiat. Transf. **18**, 533 (1979).
20. D. Chornay *et al.*, J. Phys. B **17**, 3173 (1984).
21. S. Federman *et al.*, Astrophys. J. **401**, 367 (1992).
22. S. Wu *et al.*, Phys. Rev. A **51**, 4494 (1995).
23. R.C.G. Ligtenberg *et al.*, Phys. Rev. **49**, 2363 (1994).
24. N.D. Gibson, J.S. Risley, Phys. Rev. A **52**, 4451 (1995).
25. M.C.E. Huber, R.J. Sandeman, Rep. Prog. Phys. **49**, 397 (1986).
26. D.N. Stacey, J.M. Vaughan, Phys. Lett. **11**, 105 (1964).
27. T. Sawada *et al.*, Phys. Rev. A **4**, 193 (1971).
28. C.M. Lee, K.T. Lu, Phys. Rev. A **8**, 1241 (1973).
29. C.M. Lee, Phys. Rev. A **10**, 584 (1974).
30. D.L. Lin *et al.*, Phys. Rev. A **16**, 589 (1977).
31. E.N. Avgoustoglou, D.R. Beck, Phys. Rev. A **57**, 4286 (1988).
32. I.M. Savukov, J. Phys. B **36**, 2001 (2003).

33. I. Irimia, C. Fischer, J. Phys. B **37**, 1659 (2004).
34. D.J. Chornay *et al.*, J. Phys. B **17**, 3173 (1984).
35. I.M. Savukov, H.G. Berry, Phys. Rev. A **67**, 032505 (2003).
36. M. Chenevier, P.A. Moskowitz, J. Phys. B **35**, 401 (1974).
37. M. Aymar, M.G. Schweighofer, Physica **67**, 585 (1973).
38. NIST atomic spectra database, <http://physics.nist.gov/cgi-bin/AtData/main.asd>.
39. X. Husson, J.P. Grandin, J. Phys. B **11**, 1393 (1978).
40. G. Binet, X. Husson, J.P. Grandin, J. Phys. Lett. **44**, 151 (1983).
41. H. Abu Safia *et al.*, J. Phys. B **14**, 3363 (1981).
42. A. Messiah, *Quantum Mechanics*, Vol. **2** (North Holland Publishing Co., Amsterdam, 1965) p. 927.
43. I.P. Grant, J. Phys. B **7**, 145 (1974).
44. R.D. Cowan, *The Theory of Atomic Structure and Spectra* (University of California, Berkeley, CA, 1981).
45. J.D. Jackson, Am. J. Phys. **70**, 917 (2002).
46. G.S. Adkins, Phys. Rev. D **36**, 1929 (1987).
47. F.A. Babushkin, Opt. Spectrosc. **13**, 77 (1962).
48. F.A. Babushkin, Acta Phys. Pol. **25**, 749 (1964).
49. F.A. Babushkin, Opt. Spectrosc. **19**, 1 (1965).
50. A.R. Edmonds, *Angular Momentum in Quantum Mechanics* (Princeton University Press, 1996).
51. A.J. Akhiezer, V.B. Berestetskii, *Quantum Electrodynamics* (Interscience, New York, 1965) Chapt. 8, Sect. 50.2.
52. N.C. Pyper, I.P. Grant, N. Beatham, Comput. Phys. Commun. **15**, 387 (1978).
53. I. Lindgren, A. Rosen, Case Stud. At. Phys. **4**, 93 (1975).
54. I. Lindgren, A. Rosen, Case Stud. At. Phys. **4**, 197 (1975).
55. J.P. Declaux, Comput. Phys. Commun. **9**, 31 (1975).
56. C. Fischer, Phys. Scr. T **134**, 014019 (2009).
57. J. Ekman *et al.*, Atoms **2**, 215 (2014).
58. P. Jönsson *et al.*, Comput. Phys. Commun. **184**, 2197 (2013).
59. C. Froese Fischer, Comput. Phys. Commun. **1**, 151 (1969).
60. E.R. Davidson, J. Comput. Phys. **17**, 87 (1975).
61. M. Aymar, Physica **57**, 178 (1971).
62. R.M. Sternheimer, Phys. Rev. **146**, 140 (1966).
63. R.M. Sternheimer, Phys. Rev. **164**, 10 (1967).
64. C. Froese Fischer *et al.*, *Computational Atomic Structure* (Institute of Physics Publishing, Bristol, 1997) p. 10.
65. C. Froese Fischer, X. He, Can. J. Phys. **77**, 177 (1999).
66. G. Tachiev, C. Fischer, J. Phys. B **32**, 5808 (1999).
67. C. Froese Fischer *et al.*, At. Data Nucl. Data Tables **92**, 607 (2006).
68. M. Aymar *et al.*, Nucl. Instrum. Methods **90**, 137 (1970).
69. W.L. Wiese *et al.*, *Atomic Transition Probabilities*, Volume **II**, *Sodium through Calcium*, NSRDS-NB 22 (Govt. Printing Office, Washinton D.C., 1969).
70. R.S. Knox, Phys. Rev. A **128**, 375 (1958).
71. W.F. Chan *et al.*, Phys. Rev. A **46**, 149 (1992).
72. S. Tsurubuchi *et al.*, J. Phys. Soc. Jpn. **59**, 497 (1990).
73. G. Takayanagi *et al.*, Phys. Rev. A **38**, 1240 (1988).
74. N. Avgoustoglou *et al.*, Phys. Rev. A **57**, 4286 (1988).
75. U. Hahn, N. Schwentner, Chem. Phys. **48**, 53 (1980).
76. S.R. Federman *et al.*, Astrophys. J. **401**, 367 (1992).
77. D. Lang *et al.*, *6th International Colloquium On Atomic Spectra and Oscillator Strengths*, edited by J.B. Tatum (University of Victoria, Victoria, 1998) p. 90.
78. J. Geiger, Phys. Lett. A **33**, 351 (1970).
79. O. Zatsarinny, K. Bartschat, J. Phys. B **39**, 2145 (2006).
80. G.M. Lawrence, Phys. Rev. **175**, 40 (1968).
81. W.L. Wiese *et al.*, *Atomic Transition Probabilities*, Volume **I**, NSRDS. NBS 22 (Govt. Printing Office, Washinton D.C., 1966).
82. R.L. Burnham, R.C. Isler, Bull. Am. Phys. Soc. **15**, 213 (1970).
83. D.J.G. Irwin *et al.*, Nucl. Instrum. Methods **110**, 111 (1973).
84. E.L. Lewis, Proc. Phys. Soc. **92**, 817 (1967).
85. W.L. Wiese *et al.*, Phys. Rev. **39**, 2461 (1989).
86. R.S.F. Chang, D.W. Setser, J. Chem. Phys. **64**, 3885 (1978).
87. R.A. Lilly, J. Opt. Soc. Am. **66**, 245 (1976).
88. K. Musiol *et al.*, J. Quant. Spectrosc. Radiat. Transf. **29**, 321 (1983).
89. J.B. Shumaker Jr., C.H. Poneno, J. Opt. Soc. Am. **57**, 8 (1967).
90. M. Aymar, M.G.M. Coulombe, At. Data Nucl. Data Tables **21**, 537 (1978).
91. R. Kurucz, <http://kurucz.harvard.edu/atoms/1800/life1800.dat>.
92. H. Abu Safia, J. Margerie, J. Phys. B **16**, 927 (1983).
93. C.E. Moore, *Atomic Energy Levels*, NBS Circular No. 467, Volume **I** (US Govt Printing Office, Washinton, DC, 1949).
94. C.E. Moore, *Atomic Energy Levels*, NBS Circular No. 467, Volume **2** (US Govt Printing Office, Washinton, DC, 1952).
95. C.E. Moore, *Atomic Energy Levels*, NBS Circular No. 467, Volume **I** (US Govt Printing Office, Washinton, DC, 1959).
96. S. Liberman, J. Phys. (Paris) **30**, 53 (1969).
97. S. Liberman, Physica **69**, 598 (1973).